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Survey: application and analysis of generative adversarial networks in medical images

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Abstract
Generative Adversarial Networks (GANs) have shown promising prospects and achieved significant results in medical image analysis tasks. This article provides a comprehensive review of recent research on GANs and their variants in medical applications, including tasks such as image synthesis, segmentation, classification, detection, denoising, reconstruction, fusion, registration, and prediction. We summarize and analyze the reviewed literature, with a focus on model framework design,dataset sources, and performance evaluation. Our research findings are presented in the form of tables. In the end,article discusses open challenges and directions for future research.

Keywords Generative adversarial networks · Deep learning · Medical images · GAN survey · Medical image analysis

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1 Introduction

Considering the progressive advancements in medical imaging and computer technology the automated analysis of medical images has become an indispensable tool for medical research, clinical disease diagnosis, and treatment, Typically, doctors or researchers use the relevant information provided by medical images to analyze, monitor, and plan treatment for specific internal organs. In the early years, medical imaging analysis relied mainly on traditional AI (artificial intelligence) approaches and ML (machine learning) methods, which required accurate feature annotations for images.Nevertheless,the process of manually annotating medical images by professionals is costly and time-intensive,and these features belong to a specific application and cannot be generalized, which significantly restricts the utilization of related models in this domain. With the advent of deep learning (DL) and convolutional neural networks (CNN), image analysis algorithms have reached a new starting point, as they can automatically learn important features in the data distribution without the need for specialized doctors to perform manual feature extraction and annotation. It enables non-medical researchers to effectively use deep models to carry out medical image analysis and research work, and has achieved a series of good results. However, the training of these algorithms necessitates a substantial quantity of accessible medical data, and the shortage of existing medical images has hindered further development in the field of medicine.

Until recently, the emergence of generative adversarial networks (GANs) with adversarial training mechanisms has further promoted the research boom in the field of medical images. Because GANs can generate realistic images by simulating the distribution of real clinical data, they have opened up a new chapter in medical image generation modeling and distribution learning.This review provides an overview of the most recent research conducted between 2021 and early 2023, focusing on the applications of GAN-based architectures in the field of medical image processing (Kazeminia et al. 2020; AlAmir and AlGhamdi 2022). We categorized the review papers into ten distinct categories based on the following applications: synthesis, segmentation, classification, detection, denoising, reconstruction, fusion, registration, prediction (as shown in Fig 1), and image enhancement (as shown in Fig 2). GAN-based techniques have been employed to various medical imaging modalities and disease categories,including CT (computed tomography), X-ray, PET (positron emission tomography) and MRI (magnetic resonance imaging), ultrasound, OCT (optical coherence tomography), vascular, dermatoscopic, and microscopic examinations. Disease categories include brain, eye, thyroid, breast, lung, cardiology, abdomen (kidney, liver, prostate), spine, and cell.

We found the necessary papers by searching for the keywords “medical images” and “GAN” or “Generative Adversarial Network” on Google Scholar. To facilitate reading and quick access, we created an information table on the papers, which includes the application framework, loss function, image modality, dataset, performance, and whether it contains code. In addition, we further discussed the advantages and disadvantages of these applications and briefly indicated future research directions.

This comprehensive review article encompasses various applications and categories of GANs in the domain of medical imaging, including 214 most relevant papers. The structure of the following sections is as follows: In Chapter 3, the general concepts of GANs for medical image applications, their conditional variants, and well-known improved frame-

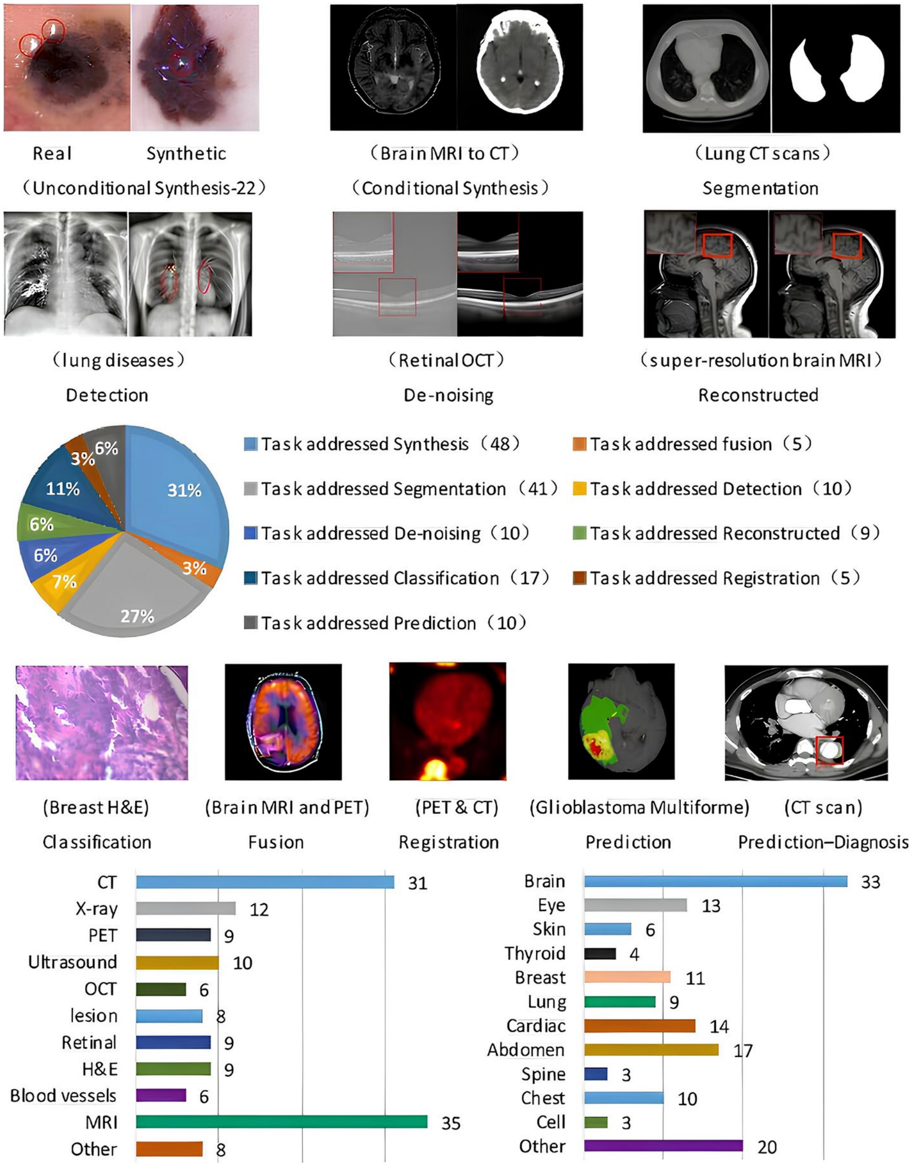

Fig. 1 The distribution of papers and visual illustrations of GAN functionality among various applications is depicted in the pie chart. The provided visual examples are sourced from papers, specifically focusing on conditional synthesis (Hassan et al. 2022), unconditional synthesis (Chen et al. 2022c), segmentation (Xun et al. 2022), detection (Li and Li 2023), de-noising (Kumar et al. 2023), reconstruction (Sun et al. 2022c), classification (Jeong et al. 2022), fusion (Zhou et al. 2023), registration (Zheng et al. 2022), prediction-diagnosis (Qi et al. 2022), and prediction (Xu et al. 2022a)

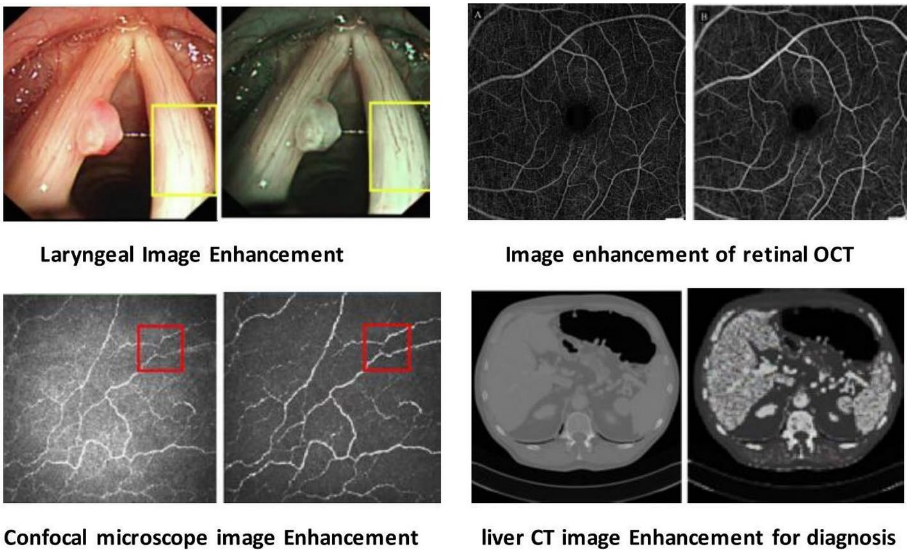

Fig. 2 The figure demonstrates the application of GAN-based techniques in enhancing medical images to assist in disease diagnosis. The visual examples provided are from respective papers, showcasing the effectiveness of image enhancement techniques in the following aspects: Pan et al. (2023) preserving the structure of Laryngeal Colonoscope images while enhancing detail prominence and image quality; Yuan et al. (2022) improving the quality of wide-field retinal optical coherence tomography angiography (OCTA) images; Yu et al. (2023) generating confocal microscope images with more complete texture structures and uniform illumination distribution; and Khan et al. (2023) preprocessing medical images to retain more liver CT structure details, for malignant and normal diagnosis of liver cancer

works based on them are introduced. In Sect. 4, the different functions and contributions of GANs in medical image processing applications (4.1-synthesis, 4.2-segmentation, 4.3-classification, 4.4-detection, 4.5-denoising, 4.6-reconstruction, 4.7-fusion, 4.8-registration, 4.9-prediction, and 4.10-image enhancement) are described. Section 5 provides conclusions about the advantages and disadvantages of research methods using GANs in medical image processing, as well as future research directions.

2 Opportunities for medical image assessment

One of the important auxiliaries, deep learning techniques based on supervised learning have become important auxiliary tools in computer vision and analysis of medical images. However, the effectiveness of these techniques relies heavily on the presence of abundant labeled training datasets. In medical image analysis tasks, the scarcity of annotated image data is a common problem due to patient privacy protection, high collection costs, and time-consuming manual labeling. Additionally, there are many rare or uncommon diseases in the medical field, which leads to extremely imbalanced classes in private or open-source datasets. All these factors severely affect the effectiveness of deep learning algorithms in medical image analysis tasks. With the continuous development of computer technology, the attention devoted to GAN models by researchers has been on the rise. This is mainly attributed to the ability of GANs to synthesize realistic images by learning the feature distribution of real images, effectively alleviating the limitations caused by data shortage. In short, GANs possess the capacity to produce images that closely resemble real images. As the potential of GANs is gradually being explored, researchers are actively exploring whether they can better solve these existing problems.

Usually, GAN models are roughly divided into two forms: unconditional and conditional. Various variant forms of GAN are widely used in medical image analysis. Early GAN was proposed as an unsupervised (unconditional) generative network framework. In image synthesis, the unconditional generative framework typically maps random noise to realistic target images (see Sect. 4.1.1). Later, variants conditioned on prior knowledge were proposed, where image feature labels (such as class labels, attributes, the image itself) were added as inputs to the model, rather than just relying on random noise to generate target images (see Sect. 4.1.2).

In the medical domain, obtaining a substantial quantity of annotated data is typically challenging and expensive, primarily due to constraints related to time and resources. However, acquiring a substantial quantity of unlabeled dataset is comparatively effortless. This has effectively assisted the progress of semi-supervised DL techniques in some previous studies. Models trained on both annotated and unannotated data are widely used in classification, segmentation, fusion, and denoising models. For example, some research proposes an extended network based on GANs, which employs a limited quantity of annotated data to train transfer learning models for classifying a large amount of unlabeled image datasets (Wang et al. 2021b) (see Sect. 4.3). The segmentation of medical images is commonly an indispensable necessity for diverse applications, including the detection of diseases, analysis, treatment, and classification. Initial medical image segmentation approaches were typically based on region, edge, classification, and traditional machine learning algorithms. Although these methods achieved good results to some extent, performance is still affected by weak edges, false edges, and noise when conducting segmentation tasks on medical images. Meanwhile, selecting appropriate thresholds is also very difficult. With the exponential growth of DL technology, GANs have largely solved the above problems (see Sect. 4.2). In addition,in terms of disease diagnosis, using a single-mode to diagnose diseases becomes a dilemma in some fields due to the lack of information. GAN-based image fusion technology is used to solve this problem, aiming to generate organ fusion images with significant features and improved quality (see Sects. 4.7), which contain information from multiple modal images, facilitating diagnosis and segmentation (Liu et al. 2022c; Abirami et al. 2022). The most basic principle of diagnosis is to be able to identify abnormalities in the marked lesion images that correspond to normal appearance images. Therefore, the utilization of GANs for generating abundant labeled training data from unlabeled data can significantly aid physicians in providing prompt patient care and enhance the efficacy of disease diagnosis (see Sect. 4.4).

In the domain of deep learning investigation, image transformation between different domains is another relevant issue that impedes development. Such transformations often exhibit poor generalization ability in medical imaging, resulting in blurry images. However, GAN-based approaches have demonstrated significant promise in the translation of images across various modalities, and with the maturity of GAN technology, it has eased many complex issues in image denoising and achieved good results (see Sect. 4.5).Currently, due to the long-term development of GAN models and their ability to synthesize realistic images, they have also provided new opportunities for medical image reconstruction or registration.

For example, GANs have the ability to reconstruct high-resolution images from low-resolution counterparts while simultaneously reducing artifacts (see Sect. 4.6),and GAN has been increasingly utilized by researchers to extract better registration mappings (see Sect. 4.8).

In addition, GAN also plays a crucial role as a clinical prediction model framework in the medical field. This mainly involves using fully or (semi/non-)parametric mathematical models to calculate the probability of an individual having a certain disease or the possibility of a certain outcome in the future (see Sect. 4.9). The most crucial aspect for enhancing the accuracy of disease diagnosis is the reliance on high-quality medical images. Image enhancement technology plays a vital role in this regard by generating medical images with richer details and higher quality (see Sect. 4.10). In the field of medical-assisted diagnosis, image enhancement technology plays a pivotal role, providing support for image preprocessing.

3 Foundational GAN models utilized in medical applications

Within this section, we will provide an overview of GANs and their conditional variants, as well as some well-known improvement frameworks built upon them. These variants include conditional GANs (cGAN), DCGAN, Progressive Growing GAN (PG-GAN), CycleGAN, WGAN (Wasserstein GAN), and its gradient penalty-incorporated version, WGAN-GP. Additionally, we will discuss the advancements made by CoGAN, ProGAN, SRGAN, BigGAN,and SAGAN, which have been widely applied in medical image processing tasks.

3.1 GAN

Generative Adversarial Networks Goodfellow et al. (2020) is currently the most prominent unsupervised generative algorithm consisting of two models, a G (generator) and a D (discriminator), which continuously optimize through a game (refer to Fig 3). Typically, we use GAN to generate realistic images. In medical applications, GAN is often used to generate required medical images. During the training process of the GAN framework, it contains a real training set X labeled as $P_{data}(X)$ and random noise Z labeled as $P_z(Z)$. The model initializes the generator parameter as θ_G and discriminator parameter as θ_D. The goal of G is to sample M real samples $x_1, x_2, x_3, \ldots, x_m$ from the data distribution $P_{data}(X)$ and M noise samples $z_1, z_2, z_3, \ldots, z_m$ from the prior distribution $P_z(Z)$, subsequently utilize the sampled data as the input for G to synthetic samples $\hat{x}_1, \ldots, \hat{x}_m$ denoted as $\hat{X}$. After fixing the generator, we use the gradient ascent strategy to train D to more effectively discern between authentic and artificial samples. In essence, D acts as a binary classifier that assigns a value of $D(x) = 1$ to genuine samples and $D(\hat{x}) = 0$ to fabricated samples. After

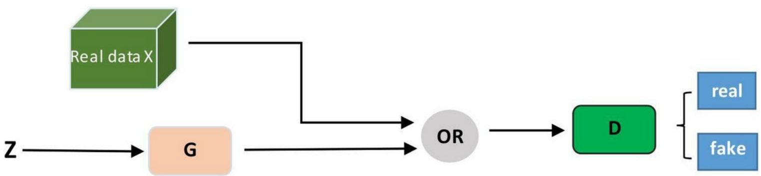

Fig. 3 Basic architecture diagram of GAN. 'OR' represents a selector that passes real data and generated data to the discriminator. It signifies that the discriminator receives either real data or generated data

training D for multiple iterations, we optimize G using a smaller learning rate and gradient descent strategy. After multiple updates of G and D training, the desired outcome we aim for is that G produces synthetic samples $\hat{x}$ that are indistinguishable by D, denoted as $\hat{x} = G(z;\theta_G) \sim \hat{X}$, and the final classification accuracy of D is 0.5. Mathematically, the value function $V(G, D)$ can be utilized to depict the two game processes within GAN, and the issue is converted into addressing the minimax problem of this value function. As shown in Eq. (1):

$$\begin{aligned}\min_G \max_D V(G, D) = \mathbb{E}_{x\sim p_{data}(x)}[\log D(x)] \\ + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))]\end{aligned} \tag{1}$$

During the training process, G is trained to generate realistic samples that can deceive D by minimizing $\log(1 - D(G(z)))$, making D believe that the synthesized samples are authentic. On the flip side, D is trained to minimize the BCE(binary cross-entropy) loss function in order to maximize the likelihood of accurately classifying counterfeit and genuine data samples.

In theory, GAN has a strong foundation for generating synthetic data that approximates real data. However, in practice, it has been shown to be difficult to train due to various issues. These include convergence difficulties (Salimans et al. 2016), vanishing or exploding gradients (Arjovsky and Bottou 2017), overfitting (Yazıcı et al. 2018), and mode collapse (Radford et al. 2015a). Convergence difficulties occur when the training process does not reach a stable equilibrium between G and D. Vanishing or exploding gradients occur when the gradients in the training process become too small or too large, respectively, making it difficult to update the parameters. Overfitting happens when the model becomes excessively intricate and perfectly matches the training data, leading to subpar performance in terms of generalization. Mode collapse occurs when G generates only a limited set of samples that do not cover the full distribution of the real data.

To address these problems, many researchers have proposed various extensions and subclasses of GAN. We will discuss some of them below.

3.2 cGAN

The cGAN (conditional Generative Adversarial Network) Mirza and Osindero (2014) is an extension of the original GAN that allows the generation of images based on additional auxiliary information in addition to the noise vector. In the initial GAN, the input data consists of a latent variable z randomly selected from a Gaussian distribution, while the cGAN has two input data, namely the latent variable z and the control condition y (Fig. 4). Specifically, the cGAN model can implement conditional constraints by adding an additional input layer to simultaneously input the additional information y into both generator G and discriminator D. y can take various forms of auxiliary information, for instance features, class labels, or even a numerical value. The G combines the initial noise input $P_z(z)$ with y to produce the generated output represented as $\hat{x} = G(z|y)$. The D in the model evaluates if the generated image $\hat{x}$ created by G satisfies the specified condition y. In simpler terms, the objective of $D(x|y)$ is to make the pairing between real samples $P_{data}(x)$ and label y close to 1, and to

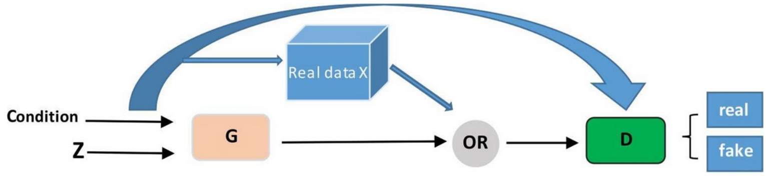

Fig. 4 Basic architecture diagram of cGAN

make the pairing between generated samples $G(z|y)$ and label y close to 0. For real samples with mismatched label y, the output of D is close to 0.

The objective function of conditional generative adversarial network training is almost the same as that of the original GAN, both of which are a process of maximizing and minimizing. The loss function of cGAN is defined as follows:

- z is the latent variable randomly selected from a Gaussian distribution.
- x represents real data samples from the dataset.
- y can take various forms of auxiliary information, such as features, class labels, or even numerical values.

$$\begin{aligned} L_{cGAN}(G, D) = & \mathbb{E}_{x,y}[\log D(x, y)] \\ & + \mathbb{E}_{x,z}[\log(1 - D(x, G(x, z)))] \end{aligned} \tag{2}$$

The first term in the above Eq. (2) represents the input to D, comprising the image x along with the condition y. When the label corresponds to the real image's label, the desired output is to be as close to 1 as possible; otherwise, it should be near 0. The second component consists of the expression $D((G(z|y))|y)$, which is used to optimize D and G using generated samples and labels y (optimizing D during D training and optimizing G during G training).The authors of Mirza and Osindero (2014) conducted experiments on image data and found that cGAN improved the stability of the model during training and significantly improved the detailed features in the generated target images. Another successful variant of cGAN is Pix2Pix, proposed in Isola et al. (2017), which is considered the pioneer of image translation. Pix2Pix is a framework for translating one image to another based on U-Net (Ronneberger et al. 2015) and PatchGAN (Isola et al. 2017). It combines L1 distance and adversarial loss, as well as other novel designs in G and D, to perform well in various image translation tasks.

The authors selected the U-Net framework as the model generator because the skip connections help improve the overall consistency of the generated images. The PatchGAN, which follows a fully convolutional architecture, which was selected as the model's discriminator to tell apart real and fake detailed image data. In addition, unlike the original GAN, Pix2Pix requires paired image datasets as input and output data.

In general, cGAN is an enhanced version of the original GAN that allows for generating images based on specific conditions. It has numerous practical uses, such as transforming images, creating new images, and editing existing images, and continues to be an actively explored field of study.

3.3 DCGAN

To tackle the issue of instability in the basic GAN architecture, Deep Convolutional GANs (DCGANs) were developed (Radford et al. 2015b). These enhanced models aimed to provide a more stable framework for generating realistic images. DCGANs replace fully connected layers with convolutional layers and use transpose convolution with strided convolution to more effectively extract image features and enhance the quality of synthesized images. Additionally, DCGANs employ concepts such as LeakyReLU activation functions and batch normalization in the discriminator to prevent gradient sparsity and further improve model stability during training. Despite these improvements, mode collapse remains a potential issue in DCGANs. The loss functions used in DCGANs are similar to those in the original GAN formulation but are adapted to the convolutional architecture. The DCGAN loss is defined in Eqs. (3), (4), (5):

$$\mathcal{L}_D = -\mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] - \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}}[\log(1 - D(G(\mathbf{z})))] \tag{3}$$

$$\mathcal{L}_G = -\mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}}[\log D(G(\mathbf{z}))] \tag{4}$$

In practice, an alternative loss function for the generator is often used to avoid saturation of the gradients. This alternative loss function aims to maximize the probability of the discriminator being mistaken, which is equivalent to minimizing:

$$\mathcal{L}'_G = \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}}[\log(1 - D(G(\mathbf{z})))] \tag{5}$$

where $\mathbf{x}$ represents real images from the data distribution p_{data}, and $\mathbf{z}$ represents random noise from the noise distribution $p_{\mathbf{z}}$.

3.4 PGGAN

The DCGAN architecture was an improvement over the basic GAN, but it still had limitations when it came to generating high-resolution images. Specifically, the details of the images were often lost, and the discriminator could easily identify the generated images as fake, which made training difficult. To address this issue, a progressive network called Progressive Growing Generative Adversarial Networks (PGGAN) was proposed in Demir and Unal (2018). PGGAN aims to gradually increase the complexity of the network as it trains, enabling the model to improve its ability to generate higher-quality images as training progresses. While PGGAN has been successful in generating high-quality images, there are still some limitations. Because PGGAN generates images by simply adding layers without any control at each level, it is difficult to understand what the network has learned at each resolution level. This lack of control over the image generation process limits the network's overall effectiveness. The loss functions used in PGGAN can refer to the methods used in DCGAN. PGGAN's primary innovation lies in its training process, which progressively increases the network's complexity. However, its basic loss functions remain similar to those in GAN and DCGAN.

3.5 CycleGAN

For image translation between two different domains X and Y, a model that can create a connection between the samples in space X and space Y, by analyzing the characteristics of the images in samples X and Y, the aim to uncover the hidden connection between the two sets of samples. CycleGAN (Cycle-Consistent GAN), created by Zhu et al. (2017), is a model that can achieve this image translation task.

The actual framework structure of CycleGAN uses two GANs to perform the translation task (Fig. 5). The primary goal of the first GAN is to understand how to transform images from X to Y. This transformation is represented by $G_{X\to Y}$, which acts as the generator in the GAN framework. It takes an image x from X and produces a corresponding image $G_{X\to Y}(x)$ in Y, and vice versa. The second GAN aims to learn the mapping from Y to X (denoted as $G_{Y\to X}$). $G_{Y\to X}$ can convert an image y in Y to an image $G_{Y\to X}(y)$ in X, and vice versa. Two discriminators, D_y and D_x, are used to train $G_{X\to Y}(x)$ and $G_{Y\to X}(y)$, respectively. The two discriminators D are used to discriminate between the real and synthetic images created by the generators $G_{X\to Y}$ and $G_{Y\to X}$, respectively, thereby forming an adversarial generative network.

To ensure that the image translated from X to Y domain can also be translated back, the authors combine the two GANs together and use cycle consistency loss. This loss function minimizes the difference between X/Y images and $G_{Y\to X}(G_{X\to Y}(x))$ images, and also the difference between Y/X images and $G_{X\to Y}(G_{Y\to X}(y))$ images. This ensures that the model doesn't transform all images in domain X to the same image in domain Y or vice versa. The CycleGAN loss is defined in Eqs. (6), (7):

$$\begin{aligned}\mathcal{L}(G_{X\to Y}, G_{Y\to X}, D_X, D_Y) = {} & \mathcal{L}_{GAN}(G_{X\to Y}, D_Y, X, Y) \\ & + \mathcal{L}_{GAN}(G_{Y\to X}, D_X, Y, X) \\ & + \lambda \mathcal{L}_{cycle}(G_{X\to Y}, G_{Y\to X})\end{aligned}, \tag{6}$$

$$\begin{aligned}\mathcal{L}_{cycle}(G_{X\to Y}, G_{Y\to X}) = {} & \mathbb{E}_{x\sim p_{data}(X)}[\|G_{Y\to X}(G_{X\to Y}(x)) - x\|_1] \\ & + \mathbb{E}_{y\sim p_{data}(Y)}[\|G_{X\to Y}(G_{Y\to X}(y)) - y\|_1]\end{aligned}, \tag{7}$$

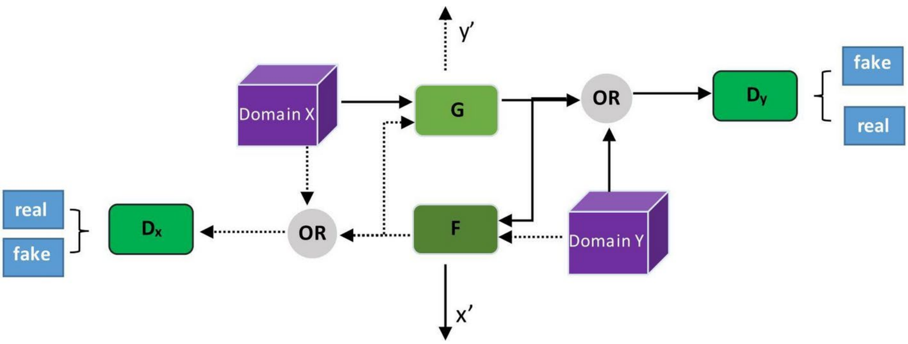

Fig. 5 Basic architecture diagram of CycleGAN

where $\mathcal{L}_{GAN}(G, D, X, Y)$ is the standard GAN loss, and $\mathcal{L}_{cycle}(G_{X\to Y}, G_{Y\to X})$ is the cycle consistency loss. The parameter λ determines how much weight is given to each loss term in the overall objective. $\mathbb{E}_{x\sim p_{data}(X)}$ and $\mathbb{E}_{y\sim p_{data}(Y)}$ denote the expectations with respect to the data distributions X and Y, respectively. $\|\cdot\|_1$ denotes the L_1 norm, used to measure pixel-level differences.

3.6 Wasserstein-GAN

Theoretically, the instability in GAN training stems from the inadequacy of cross-entropy or JS divergence as a measure for quantifying the difference between the distributions of generated and real data. This leads to challenges such as vanishing gradients and difficulty in optimizing saddle points. When the discriminator becomes overly effective, it provides weak gradients to the generator, hampering its optimization. Conversely, if the discriminator is too weak, it fails to provide meaningful feedback, hindering effective learning by the generator. Arjovsky et al. (2017) Proposed WGAN (Wasserstein-GAN),which focuses on enhancing GAN through modifications to the loss function, resulting in improved performance, particularly in fully connected layers. WGAN utilizes the Wasserstein distance to determine the dissimilarity between the distribution of generated data and the distribution of actual data. The model uses Wasserstein distance as a better divergence measure to avoid gradient disappearance, theoretically solving the problem of unstable training. It also addresses the issue of mode collapse, making the generator's results more diverse. However, WGAN sometimes has problems such as low quality of synthesized samples and difficult training convergence. Gulrajani et al. (2017) Proposed an improved version of WGAN, WGAN-GP, which mainly improves the Lipschitz continuity constraint. Because weight clipping was directly used to clip the weights to a certain range [− 0.01, 0.01], the modeling ability of the model was weakened, and there was gradient disappearance or explosion. Therefore, WGAN-GP introduced a gradient penalty term, which slightly outperformed WGAN in terms of training speed and sample quality. Since gradient penalty is applied to every sample in each batch, the authors replaced batch normalization in the discriminator with layer normalization. The loss functions for WGAN and WGAN-GP are defined in Eqs. (8) and (9), respectively:

$$\mathcal{L} = \mathbb{E}_{x\sim p_{data}(x)}[D(x)] - \mathbb{E}_{z\sim p_z(z)}[D(G(z))] \quad , \tag{8}$$

$$\mathcal{L} = \mathbb{E}_{x\sim p_{data}(x)}[D(x)] - \mathbb{E}_{z\sim p_z(z)}[D(G(z))] + \lambda \mathbb{E}_{\hat{x}\sim p_{\hat{x}}}[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2] \quad , \tag{9}$$

where:

- $x \sim p_{data}(x)$: Real sample x drawn from the true data distribution $p_{data}(x)$.
- $D(x)$: Score given by the discriminator D to the real sample x.
- $z \sim p_z(z)$: Noise vector z drawn from the noise distribution $p_z(z)$.
- $G(z)$: Generated sample $G(z)$ obtained by mapping the noise vector z through the generator G.
- $D(G(z))$: Score given by the discriminator D to the generated sample $G(z)$.
- λ : Coefficient of the gradient penalty term, controlling the strength of the gradient penalty.

- $\hat{x} \sim p_{\hat{x}}$: Sample $\hat{x}$ drawn from the interpolation between the real and generated data distributions.
- $\|\nabla_{\hat{x}} D(\hat{x})\|_2$: L2 norm of the gradient of the discriminator D output with respect to the interpolated sample $\hat{x}$.
- $(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2$: Gradient penalty term, enforcing the L2 norm of the gradient to be close to 1, ensuring the discriminator D is 1-Lipschitz continuous.

The WGAN-GP loss function adds a gradient penalty term to the original WGAN loss, replacing the weight clipping method to enhance model stability and performance. The gradient penalty term is computed by forcing the L2 norm of the gradient to be close to 1 for each sample in the batch, ensuring the Lipschitz continuity of the discriminator.

3.7 CoGAN

CoGAN (Coupled Generative Adversarial Networks) Liu and Tuzel (2016) aimed at learning the correspondence between multiple domains, allowing samples generated in one domain to match those in another domain. It consists of two generators and two discriminators, with each generator responsible for one domain and each discriminator corresponding to its generator. The major advantage of CoGAN lies in its ability to achieve cross-domain image transformation without requiring paired training data, making it highly practical in various real-world applications. The working principle of CoGAN involves training two generators and two discriminators to ensure that data from different domains can have similar representations in the latent space. During training, the generator attempts to produce samples that can deceive the discriminator, while the discriminator tries to distinguish between real and generated samples. Through alternating training of the generator and discriminator, CoGAN achieves cross-domain image transformation, enabling generated images in one domain to resemble real images in another domain.

CoGAN's loss function typically consists of two parts: generator loss and discriminator loss. The generator loss indicates that the generated samples can deceive the corresponding discriminator, while the discriminator loss indicates that the discriminator can accurately differentiate between real and generated samples. The overall optimization objective is to minimize the generator loss and maximize the discriminator loss, with one balancing factor being the hyperparameter λ.

The mathematical representations of generator loss Eq. (10) and discriminator loss Eq. (11) are as follows:

$$\mathcal{L}_{\text{gen}} = \mathbb{E}_{x_A \sim p_{\text{data}}(x_A)}[\log D_B(G_A(x_A))] + \mathbb{E}_{x_B \sim p_{\text{data}}(x_B)}[\log D_A(G_B(x_B))] \quad , \tag{10}$$

$$\mathcal{L}_{\text{dis}} = -\mathbb{E}_{x_A \sim p_{\text{data}}(x_A)}[\log D_A(x_A)] - \mathbb{E}_{x_B \sim p_{\text{data}}(x_B)}[\log D_B(x_B)] \quad , \tag{11}$$

where G_A and G_B are the generators for domain A and B respectively, D_A and D_B are the discriminators for domain A and B respectively, and $p_{\text{data}}(x_A)$ and $p_{\text{data}}(x_B)$ are the true data distributions for domain A and B respectively.

Although CoGAN performs well in many tasks, it also has some limitations. For example, for complex data distributions such as natural images, the learning process of CoGAN may become more challenging, requiring more training time and computational resources.

The training of CoGAN may be more difficult and unstable because it needs to learn the mapping relationships between multiple domains simultaneously and ensure alignment of samples across different domains. Additionally, CoGAN is highly sensitive to the selection of hyperparameters, requiring careful tuning to achieve good performance.

3.8 ProGAN

ProGAN (Progressive Growing of GANs) Karras et al. (2017) is a unique generative adversarial network that employs a progressive training approach to generate high-quality images by gradually increasing the network's depth and resolution.Traditional deep learning models often establish the entire network structure at once during training and then proceed with training. However, ProGAN adopts a progressive training method. It starts training with low-resolution images and progressively increases the depth and resolution of the network. This approach allows the network to first learn the overall structure and general features of the images and then gradually refine and enhance the finer details. This progressive training method enables ProGAN to generate more realistic and authentic images, leading to significant advancements in the field of image generation.Additionally, ProGAN has the advantage of effectively avoiding the problem of mode collapse. Mode collapse occurs when a generative adversarial network starts generating large numbers of similar repetitive images instead of diverse and rich images. By gradually increasing the complexity and resolution of the network, ProGAN ensures that the network maintains diversity and richness throughout the training process, effectively mitigating the issue of mode collapse.

However, ProGAN also has some drawbacks. Firstly, it requires substantial computational resources and training time. Because ProGAN gradually increases the complexity and resolution of the network, it requires more computational resources and longer training time to complete training. Secondly, ProGAN is sensitive to network architecture and hyperparameters. Due to its complex network structure, careful adjustments to the network architecture and hyperparameters are needed to achieve the best generation results. These factors increase the difficulty and cost of using ProGAN. ProGAN uses basic loss functions similar to traditional GANs in the training of the generator and discriminator. The loss functions for the generator and discriminator are defined as follows in Eqs. (12) and (13):

$$\mathcal{L}_G = -\,\mathbb{E}_{z\sim p_z(z)}[\log D(G(z))] \tag{12}$$

$$\mathcal{L}_D = -\,\mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x)] - \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))] \tag{13}$$

- $p_{\text{data}}(x)$ represents the distribution of real data.
- $p_z(z)$ represents the noise distribution, which is usually a uniform or Gaussian distribution.
- $G(z)$ represents the fake data generated by the generator G given the noise input z.
- $D(x)$ represents the discriminator D 's judgment on the input x, outputting a value between 0 and 1, which indicates the probability that x is real data.

3.9 SRGAN

SRGAN (Super-Resolution Generative Adversarial Networks) is a type of generative adversarial network used for image super-resolution (Ledig et al. 2017). Its main objective is to generate high-resolution images from low-resolution inputs. One of the advantages of SRGAN is its ability to produce clearer and more realistic images, leading to significant advancements in the field of image super-resolution. In addition to generating high-quality images, SRGAN is also effective in handling image transformations across different scales. This means it can not only upscale low-resolution images to high-resolution ones but also perform various scale transformations, such as converting small-sized images to large-sized ones.This versatility makes SRGAN have enormous potential applications in the field of medical image processing. However, similar to ProGAN, the training process of SRGAN requires a large amount of data and computational resources, especially when dealing with high-resolution images. Furthermore, like CoGAN, SRGAN is sensitive to the choice of network architecture and hyperparameters, requiring careful tuning to achieve optimal performance. Therefore, extensive experiments and optimizations are necessary in practical applications to ensure that SRGAN achieves the expected results. SRGAN's total loss function (Eq. 14) typically consists of two parts: Content Loss (Eq. 15) and Adversarial Loss (Eq. 16).

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{content}} + \lambda \mathcal{L}_{\text{GAN}}, \tag{14}$$

$$\mathcal{L}_{\text{content}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \phi\left(G(\mathbf{I}_{\text{LR}})\right) - \phi\left(\mathbf{I}_{\text{HR}}\right) \right\|_2^2 \quad , \tag{15}$$

$$\mathcal{L}_{\text{GAN}} = -\log D(G(\mathbf{I}_{\text{LR}})) \quad , \tag{16}$$

where:

- λ balances the importance between the content loss and the adversarial loss.
- $G(\mathbf{I}_{\text{LR}})$ denotes the high-resolution image generated by the generator network.
- $\mathbf{I}_{\text{HR}}$ represents the ground truth high-resolution image.
- ϕ denotes the high-level feature maps extracted from a pretrained VGG network.
- N is the number of pixels.
- D represents the discriminator network.

3.10 BigGAN

BigGAN is a large-scale Generative Adversarial Network (GAN) Brock et al. (2018) designed to generate images with higher resolution and more diversity. One of its main advantages is its ability to produce high-resolution and diverse images, leading to significant advancements in the field of image generation. Compared to other Generative Adversarial Networks, BigGAN can generate more realistic and varied image samples, providing broader possibilities for image generation tasks. Additionally, BigGAN can generate high-quality images in a relatively short training time, making it highly practical for real-world applications. This means that users can obtain the desired image generation results more quickly, thereby improving work efficiency and productivity. BigGAN loss consists of generator (G) loss (Eq. 17) and discriminator (D) loss (Eq. 20). The generator loss function comprises adversarial loss (Eq. 18) and class loss (Eq. 19):

However, despite its significant advantages in generating high-resolution and diverse images, BigGAN also faces some challenges. Firstly, achieving high-quality image generation requires a considerable amount of computational resources and training time, which may limit its use in applications with high real-time requirements. Secondly, BigGAN is highly sensitive to the choice of network architecture and hyperparameters, requiring extensive experimentation and expertise to adjust and optimize these parameters. These challenges necessitate careful consideration and preparation when deploying and applying BigGAN.Nevertheless, with ongoing technological advancements and optimizations, it is believed that BigGAN will continue to play a significant role and contribute to further innovations and advancements in the field of image generation.

$$\mathcal{L}_{\mathrm{G}} = \mathcal{L}_{\mathrm{Gan}} + \lambda \mathcal{L}_{\mathrm{class}} \quad , \tag{17}$$

$$\mathcal{L}_{\mathrm{Gan}} = -\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}[\log D(G(\mathbf{z},\mathbf{y}))] \quad , \tag{18}$$

$$\mathcal{L}_{\mathrm{class}} = \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}[\mathrm{CrossEntropy}(C(G(\mathbf{z},\mathbf{y})),\mathbf{y})] \quad , \tag{19}$$

$$\mathcal{L}_{\mathrm{D}} = -\mathbb{E}_{\mathbf{x}\sim p_{\mathrm{data}}(\mathbf{x})}[\log D(\mathbf{x},\mathbf{y})] - \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}[\log(1 - D(G(\mathbf{z},\mathbf{y})))] \quad , \tag{20}$$

where:

- λ is a hyperparameter that balances adversarial and class losses.
- $\mathbf{z}$ is the input noise to the generator, $\mathbf{y}$ is the conditional class vector, and $G(\mathbf{z},\mathbf{y})$ is the generator's output.
- C is an auxiliary classifier used to predict image classes.
- $\mathbf{x}$ is a real image sample, $p_{\mathrm{data}}(\mathbf{x})$ is the real data distribution, and $D(\mathbf{x},\mathbf{y})$ is the discriminator's output.

3.11 SAGAN

SAGAN (Self-Attention Generative Adversarial Networks) is a variant of Generative Adversarial Networks (GANs) that introduces self-attention mechanism, Zhang et al. (2019) aiming to generate images with better global consistency. Its core innovation lies in dynamically capturing long-range dependencies between different positions in images using self-attention mechanism, thereby generating more realistic and detailed images. The generator and discriminator in SAGAN typically adopt the basic architecture of DCGAN (Deep Convolutional GAN), which includes convolutional layer, batch normalization layer, and LeakyReLU activation functions. Additionally, in SAGAN, the self-attention mechanism enables the network to effectively focus on global information during the image generation process. Specifically, the attention weights for each pair of pixels are computed to achieve this goal in SAGAN's self-attention mechanism.

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V, \tag{21}$$

where Q, K, and V are the mappings of query, key, and value, respectively, and d_k is the dimensionality of query and key. By computing the similarity between queries and keys, followed by a softmax operation, attention weights are obtained, which are then used to compute the weighted average of values.

The generator loss function of SAGAN is defined as:

$$\mathcal{L}_{\text{gen}} = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))], \tag{22}$$

where $p_{\text{data}}(x)$ represents the real data distribution, $p_z(z)$ represents the noise distribution, $D(x)$ represents the discriminator's output for real samples, and $G(z)$ represents the generator's output for noise z. The objective of the generator is to maximize the discriminator's error on generated samples to make them more realistic.

The discriminator loss function of SAGAN is given by:

$$\mathcal{L}_{\text{dis}} = -\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log D(x)] - \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))], \tag{23}$$

where the discriminator's goal is to minimize the error between real and generated samples to accurately distinguish between them.

In summary, SAGAN enhances the global consistency of Generative Adversarial Networks (GANs) by utilizing self-attention mechanisms, allowing for dynamic adjustment of relationships between pixels to generate more realistic and detail-rich images. It can generate high-resolution and diverse images in shorter training times, thereby improving the quality and diversity of image generation. SAGAN is also effective in handling image transformations across different scales, presenting extensive prospects in medical imaging applications.However, when dealing with high-resolution medical images, SAGAN faces several challenges. Firstly, it requires a large amount of medical image data for model training, which can be challenging to acquire in practical applications. Additionally, SAGAN's sensitivity to network architecture and hyperparameter selection necessitates significant computational resources during training.

4 GANs for medical image processing task

This section summarizes 10 applications of GAN-based methods in medical imaging tasks, including synthesis, segmentation, classification, detection, denoising, reconstruction, fusion, registration, prediction, and image enhancement. In each subsection, we highlight the main elements of the reviewed methods in a table, including the basic network architecture, loss function, image modality, dataset properties, evaluation performance,domain,and whether the publication provides any source code. As these papers use different metrics to evaluate different methods, we provide a brief explanation of the indicators in Table 1. Also,the abbreviations and full names of some loss functions are presented in Table 2.

4.1 Synthesis

In the early stages of its development, GAN was proposed as an unsupervised (unconditional) generative network framework. In image synthesis, such a framework can typically use random noise to generate a realistic target image. Later, a conditional GAN (cGAN) was proposed as a generative framework that incorporates prior knowledge by adding image feature labels as inputs to the model, rather than relying solely on random noise. At this point, cGANs as a supervised (conditional) generative framework contrast with unconditional GANs. Different types of medical images have been synthesized using both unconditional and conditional generative network frameworks in various ways. In the following sections, we will provide an extensive overview of these two categories of frameworks. Additionally, in the subsection on conditional generative frameworks, further classification based on imaging modality will be provided.

4.1.1 Unconditional image synthesis

In the medical imaging field, various methods based on unconditional GANs for image synthesis have emerged in recent years. The without conditions methods are summarized in Table 3, which can effectively help solve the issue of lack of clinical datasets (Abdelhalim et al. 2021; Sharmila 2021), address data imbalance, and assist in extracting deep features and latent structures of all images in the dataset. Moreover, medical image synthesis can assist in resolving persistent challenges encountered by researchers in medical image analysis who are developing deep learning models. These challenges include the limited variety of training data and the limitations of traditional data augmentation methods (Gan et al. 2022). Existing work has shown that DCGAN, as an unconditional generation framework, can be used to synthesize artificial retinal images (Devi and Kumar 2022), skin cancer lesion images (Salvia et al. 2022), or fake samples of brain tumor MRI scans (Hassan et al. 2022). In Hassan et al. (2022), the authors demonstrated through experimental results that the false brain tumor MRI data synthesized by DCGAN were perceived as real by expert doctors. In addition, PGGAN has also been used as an unconditional generation framework to synthesize Chest X-Rays (Segal et al. 2021) and CT images of vertebral compression fractures (VCF) (Sindhura et al. 2022). PGGANs, as a multi-stage generation model, generate VCF synthetic CT images with high resolution, which can effectively improve the accuracy of VCF type recognition systems (Sindhura et al. 2022). However, generating medical images using unsupervised PGGAN requires a large number of convolutional layers to obtain high-resolution images. Therefore, Waqas et al. (2022) proposed an Enhanced-GAN to generate knee MRI, where the model employs a U-net supervised deep learning architecture to assess the effectiveness of Enhanced-GAN and the synthesized data generated by PGGAN. The dice coefficient indicator obtained from experiments showed that the proposed Enhanced-GAN performed better than PGGAN. Gan et al. (2022) also developed a GAN model called HieGAN, which is designed to produce excellent synthetic knee joint images. The findings demonstrated that HieGAN performed better than the advanced PGGAN model.

Table 1 Metric explanation

Abbreviation	Metric	Application	Abbreviation	Metric	Application
WD	Wasserstein distance	syn	MMD	Maximum mean discrepancy	syn
AM	Adversarial metric	syn	MS	Mode scores	syn
SNU	Spectral normalized uniformity	syn	NCC	Normalized cross-correlation	syn
IOU	Intersection over union	syn	PCC	Pearson correlation coefficient	syn
Dice(DSC)	Dice similarity coefficient: 2 × (TP/(2TP+FP+FN))	segm, reg	RMSE	Root mean squared error	segm, deno
Jac	Jaccard	segm	PSNR	Peak signal-to-noise ratio	segm, det, deno, fus, pre, syn
IoU	Intersection over union	segm, det	SSIM	Structural similarity index	segm, det, deno, fus, pre, syn
Acc	Accuracy	segm, clas, det, pre	ASSD	Average symmetric surface distance	segm
Sen	Sensitivity	segm, clas, det, syn	PC	Percentage correct	segm
HD	Harmonic distortion	segm, fus, reg	G-mean	Geometric mean	segm
PA	Pixel accuracy	segm	LPIPS	Perceptual image patch similarity	segm, syn
Spe	Specificity	segm, clas, det	FID	Fréchet inception distance	segm, syn
F-measure		segm, det	KID	Kernel inception distance	segm, syn
PRI	Probabilistic rand index	segm	IS	Inception score	clas
VS	Vital signs assessment indicators	segm	AUC	Area under the ROC curve	clas, det, pre
Hausdorff	Hausdorff distance	segm	ROC	Receiver operating characteristics	det
F1	F1-score	segm, clas	Energy	Energy-based metric	det
PPV	Positive predictive value	segm	Trust	Trustworthiness metric	det
CSF	Cerebrospinal fluid	segm	CNR	Signal-to-noise ratio	deno
GM	Gray matter	segm	SNR	Contrast-to-noise ratio	deno
WM	White matter	segm	EPI	Edge preservation index	deno
AVD	Average velocity difference	segm, reg	SD noise	Standard deviation of noise	deno
Recall	Sensitivity	segm, clas	ENL	Equivalent number of looks	deno

Table 1 (continued)

Abbreviation	Metric	Application	Abbreviation	Metric	Application
AJI	Average jaccard index	segm	IFC	Information fidelity criterion	deno
ASD	Average surface distance	segm	MSIM	Multiscale SIM	deno
MCE	Mean classification error	segm	FSIM	Feature SIM	deno
DC	Dissimilarity coefficient	fus	NMSE	Normalized MSE	deno
SF	Saliency map fusion	fus	Q-AG	Quality assessment with gradients	fus
VIFF	Visual information fidelity	fus, syn	Q-EN	Quality assessment with entropy	fus
TBR	Target registration error	reg	NIQE	Natural image quality evaluator	fus
SUV	Structure similarity	reg	RAE	Relative absolute	pre
C-index	Concordance index	pre	MAE	Mean absolute error	pre, syn

Syn Synthesis, *Segm* segmentation, *Clas* classification, *Det* detection, *Deno* de-noising, *Rec* reconstruction, *Fus* fusion, *Reg* registration, *Pre* prediction, *TP* true positive, *TN* true negative, *FP* false positive, *FN* false negative.

4.1.2 Conditional image synthesis

4.1.2.1 CT image synthesis In the field of medicine, many diseases require the use of CT scans to aid in diagnosis and treatment. However, CT imaging exposes patients to radiation, which can cause damage to normal cells and even increase the risk of cancer. This has prompted researchers to explore the synthesis of CT images (Heng et al. 2024). Jin et al. (2021) Proposed the use of new, richer GANs models to generate realistic 3D tumors/lesions in CT images. In order to enhance the quality of synthetic CT images for accurate oncology training and validation of deep learning models, the authors introduced a new and improved generator called RicherDG. This generator incorporates richer convolutional features and dilation-gated mechanisms. Additionally, a hybrid loss function was employed during model training. The approach suggested by Jin et al. (2021) was tested and analyzed using a large dataset of CT images, including liver, kidney tumors, and lung nodules. The evaluation, both qualitative and quantitative, showed favorable outcomes. Although deep CNN-based (convolutional neural network) AI algorithms have shown significant success in CT image synthesis tasks, synthesizing high-quality CT images can also be influenced by other factors. For instance, the DL-based cGAN model mentioned in reference Bird et al. (2021) examined how factors like T2-SPACE MR sequences, patient data from various medical centers, and gender and cancer locations impact the quality of the synthesized sCT(synthetic CT) images. In the study by Bird et al. (2021), sCT generation required paired data for training, but Schaefferkoetter et al. (2021) demonstrated through experiments that a training method without paired data can also produce high-quality synthesized

Table 2 Loss explanation

Abbreviation	Full-name	Abbreviation	Full-name	Abbreviation	Full-name
Adv	Adversarial	PTMG	Pseudo targetmodality generation	Gt	Ground-truth
BCE	Binary cross-entropy	DMFE	Dual modality feature extraction	Sce	Sigmoid cross entropy
Per	Perceptual	Ce	Cross-Entropy	VD	Visual discrepancy
Id	Identity	FM(fm)	Feature matching	Cond	Conditional
RCC	Real-cycle consistent	Rec	Reconst	Uncond	Unconditional
PCC	Pseudo-cycle consistent	Sim	Similarity	RPN	Region proposal network
Cyc(CC)	Cycle-consistency	Dice	Dice similarity coefficient	TA	Temporally aware
Ref	Reference	IoU	Intersection over union	ls	Least squares
MSE	Mean squared error	EM	Earth mover	GP	Gradient profile
GDL	Gradient difference	ODM	Overlap of dice metrics	KL	Kullback–Leibler divergence
Lad	Least absolute deviation	KD	Knowledge distillation	Mce	Softmax cross entropy
Moe(MSE)(ms)	Mean square error	Em	Entropy minimization	Bc	Binary classification
Pamr	Pixel-adaptative mask refinement	NH	Norm-combined hybrid	MAE	Mean absolute error
SSIM	Structural similarity index	Fs	Feature space	Reg	Regression
Tf	Texture fidelity	Strp	Structure-preserving	Coe	Conditional entropy

CT images, albeit with additional constraints to maintain the integrity of the generated data. The authors in Liu et al. (2021b) introduced a novel approach called Multi-Cycle GAN for generating synthetic CT images from MRI (Fig. 6). This framework incorporates a module called pseudo-cycle consistency to ensure the coherence of the generated data, along with a domain control module that adds extra constraints for consistency. Furthermore, they developed a new generator called Z-Net, which enhances the precision of anatomical features.

With the development of generative networks, more and more models have made good progress in synthesizing CT images. Yoo et al. (2022) proposed to evaluate and compare the quality of sCT generated by GAN, CycleGAN, and Reference-Guided CycleGAN (RgGAN) in the prostate cancer volume modulated arc therapy (VMAT) plan. The experimental results show that the sCT generated by RgGAN exhibits the best performance in dose-metric-conservation $D_{98\%}$ and $D_{95\%}$. Typically, obtaining CT images requires the injection of contrast agents. However, patients with kidney disease and iodine allergy are

Table 3 Medical image synthesis using unconditional GANs

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Abdelhalim et al. (2021)	HAM10000 (10015 images)	Skin lesions	128×128, 256×256	SPGGAN-TTUR,CNN	Adv, Wasserstein	No	Sen	$\uparrow$ 5.6% &2.5% $\uparrow$ 13.8% &8.6%	Skin lesions synthesis
Sharmila (2021)	Kaggle (12812 images)	Chest X-ray	256×256	DCGAN-CNN	Adv	No	Acc	Enhanced classification 94.8%,96.6%, 98.5%, 98.6%	Chest X-ray synthesis
Segal et al. (2021)	ChestX-ray14 (112120 images)	Chest X-ray	1024×1024	PGGAN	Adv, WGAN-GP	Yes	FID	8.02	Chest X-ray synthesis
Gan et al. (2022)	OAI(75 images)	knee MRI	256×256	HieGAN, PGGAN	Adv, Wasserstein	No	MAE WD	0.00135 0.506	Knee MRI synthesis
Devi and Kumar (2022)	Kaggle (5654 images)	Diabetic Retinopathy Image	$4 \times 4 \times 512$	DR-DCGAN	Adv,BCE	No	Acc	Enhanced classification 98.66	DR Image synthesis
Salvia et al. (2022)	Vivo[1](61 subjects)	Skin lesions Image	50×50	DCGAN	Adv	No	FID	17.37	Skin lesions synthesis
Hassan et al. (2022)	Non-public	Brain MRI	128×128	DCGAN	Adv	No	n/a	Performance outperforms other types of GANs	Brain tumor image generations
Sindhura et al. (2022)	Non-public (375 images)	(VCF)CT	512×512	PGGAN	Adv	No	Acc	Enhanced classification 87.01%	Spinal CT synthesis
Waqas et al. (2022)	Non-public (75 images)	knee MRI	256×256	PGGAN Enhanced-GAN	Adv, WGAN	No	AM MS	3.049 1.1009	Knee MRI synthesis

1 https://www.mdpi.com/2079-9292/9/9/1503

contraindicated for injection. Therefore, Chandrashekar et al. (2023) developed a DL pipeline based on GANs that generates CTA from non-contrast images without the need for contrast agents. The authors in Liu et al. (2022b) developed a GAN model to automatically generate contrast-enhanced CT images in the abdomen and pelvis area from non-contrast CT images. A dual-path training mode is used in the model to simultaneously learn texture and structural features, which are used to accurately synthesize intensity details and residual texture structures of the CT images. Good results were obtained in experiments. There is a recent suggestion that using medical images that combine different types of information can be beneficial for radiotherapy purposes. For example, Sun et al. (2022b) proposed a novel radiotherapy treatment mode based on a triangular GAN (TGAN) model for synthesizing pseudo-medical images between multimodal datasets. The model is verified in CBCT→ CT and MRI→ CT synthesis. The results of the experiments demonstrate that the generated synthetic medical images using TGAN closely resemble real images in terms of anatomical details and dose measurements. These artificially generated medical images have great potential for applications in clinical adaptive radiotherapy. Reaungamornrat et al. (2022) Proposed the use of non-cooperative relativistic GAN to synthesize multimodal images based on anatomical and modality-specific features represented by disentangled representations. Multimodal medical images can be used for various approaches to tackle various medical diagnosis issues. However, obtaining multimodal images is often challenging due to various limitations, including high costs and concerns for patient safety. Li et al. (2022c) suggested a novel approach called Transformer-CNN-GAN (TCGAN) that combines transformer networks and CNN to generate CT images for positron emission tomography attenuation correction. Through qualitative and quantitative analysis, it was found that the proposed method surpasses existing techniques in terms of performance and quality (see Table 4).

4.1.2.2 MR image synthesis Super-Resolution (SR) for single images is about generating a high-quality, detailed image from a low-quality, blurry image. However, many current methods are limited to specific scaling factors and cannot handle different scaling factors effectively. Zhu et al. (2021) proposed a method called MIASSR (Medical Image Arbitrary Scale Super-Resolution) that uses a combination of meta-learning and GAN to enhance the resolution of medical MRI images at different scaling factors. The process of converting low-resolution MRI to high-resolution requires preserving image details, features, and structures. The authors in Xia et al. (2022) Developed a method for detecting and synthesizing MR images. Their model consists of an interpolation network that relies on a specialized conditional GAN. This network successfully retains important visual information and intricate structural elements in the synthesized MRI slices. Reference Xu et al. (2022d) introduced a new approach called Bi-MGAN, which uses a multi-generator multi-adversarial network to learn two mappings between T1-weighted MRI and T2-weighted MRI images. The experimental results demonstrate that Bi-MGAN outperforms other advanced methods in preserving both pathological features and anatomical structures. Please refer to Fig 7 for visual representation. Amirrajab et al. (2022) introduced a framework for segmenting and creating images, specifically cardiac magnetic resonance (CMR) images, using a mask-conditioned GAN. The study's findings demonstrate that incorporating synthetic

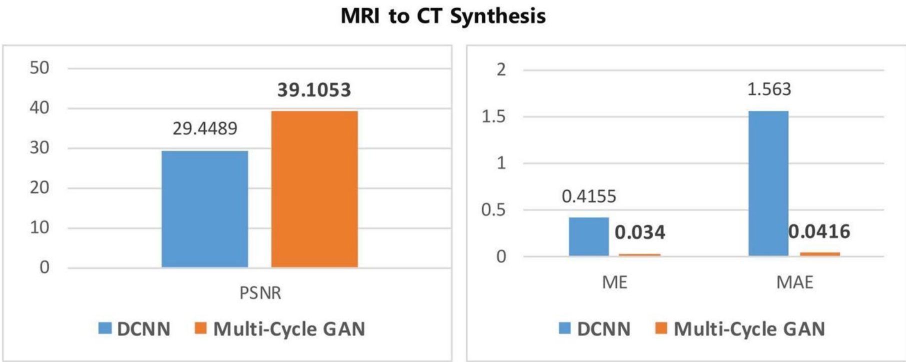

Fig. 6 Liu et al. (2021b) proposes the use of Multi-Cycle GAN to generate CT images from MRI to assist in head and neck radiotherapy. Experimental comparisons were made with DCNN generation models, revealing a significant improvement in the PSNR metric and effective reduction in ME and MAE when employing the GAN method. This demonstrates that using GAN methods outperforms traditional generation models in performance

data augmentation during training greatly enhances the accuracy of model, as evidenced by improved Dice score and Hausdorff distance measurements.

In practical situations, the amount and quality of contrast in multi-contrast MRI are restricted due to factors like scan time and patient movement. However, creating or reconstructing missing or damaged contrasts by utilizing existing high-quality contrasts can help overcome this limitation. The authors in Yurt et al. (2021) introduced a multi-stream generative adversarial network (mustGAN) for generating MR images. The mustGAN approach improves the synthesis of MR images by combining information from multiple source images through a combination of one-to-one streams and joint multi-to-one streams, resulting in superior performance.A new GAN called D2FE-GAN was created by Zhan et al. (2022) to generate multiple distinct contrast images of a patient using cross-modal MRI synthesis. This network utilizes a decoupled dual-feature representation approach for improved performance. Just like Bird et al. (2021), Conte et al. (2021) utilized paired MRI data of multiple modalities to train generative adversarial networks. Aim was to synthesize missing T1 and FLAIR MRI sequences. In the field of cross-modal MRI synthesis, numerous techniques have been developed recently. However, most of these methods focus solely on comparing pixel-level dissimilarities between the synthesized and ground truth images, neglecting important edge information. As a result, a new method called EP_IMF-GAN, proposed by Luo et al. (2021), aims to generate MRI images while preserving edges. This is achieved through an iterative multi-scale feature fusion approach, where auxiliary tasks for generating edge images are incorporated. These tasks contribute to the preservation of edge information and the improvement of latent representation features by utilizing shared encoders. Amirkolaee et al. (2022) Proposed a novel end-to-end generative adversarial network for synthesizing MR images from CT images, using an edge detection network to make the synthesized MRI tissue edges clearer. The proposed model was experimented in both paired data and unpaired data modes and compared with other image-to-image transformation methods, demonstrating better conversion performance (see Table 5).

Table 4 Medical image synthesis using conditional GANs–(CT)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Jin et al. (2021)	LiTS/KiTS (200/300 scans) LUNA (888 subjects)	3D lesion, CT iamge	512×512 512×512 512×512	FRGAN RicherDG	Adv,mask, per,style	No	n/a		Liver/kidney/ lung CT synthesis
Bird et al. (2021)	Non-public (90 subjects)	Paired MR-CT	450×450	cGAN	Adv, focal	No	Dose	D95%	Anorectal CT synthesis
Schaeffer-koetter et al. (2021)	Non-public (60 subjects)	3D MRI, CT, PET images	$96 \times 96 \times 96$	CycleGAN patchGAN	Adv, id, L_1, L_2	No	n/a	SynCT resulted in much higher PCC between CT mu values 0.885	Whole-body CT synthesis
Liu et al. (2021b)	RDD(164 subjects)	MRI, CT	384×384	multi-cycle GAN	Adv, RCC, PCC	No	MAE ME PSNR	0.0416 0.0340 39.1053	Head-and-neck CT synthesis from MRI
Yoo et al. (2022)	Non-public (113 pairs)	MRI, CT	144×144	GAN, RgGAN CycGAN	Adv Ref, cyc	No	PSNR	3620 + 0.48 36.09 + 0.44 36.31 + 0.61	Prostate CT synthesis from MRI
Chandrashekar et al. (2023)	OxAAA study(275 subjects)	NCCT, CTAs images	256×256x3	CycleGAN ConGAN	Adv, cyc, id	No	SegAcc ClasAcc	$86.1\% \pm 12.2\%$ $85.7\% \pm 10.4\%$ 93.5% 85.7%	Simulate contrast enhanced CTA images using noncontrast CTs
Liu et al. (2022b)	NLST(41589 slices) Non-public (167 subjects)	CT images	512×512	dual-path GAN-based	Adv, Per, MSE	No	PSNR SSIM MSE SNU	19.79 ± 1.17 0.65 ± 0.05 226 ± 29 0.34 ± 0.16	Abdomen/ pelvis non-contrast CT to contrast-enhanced CT synthesis
Sun et al. (2022b)	Non-public (80 subjects)	CBCT, CT, MR images	$512 \times 512 \times 30$	TGAN	Adv, cyc, label, BCE, gradient	No	PSNR	30.29 (1.18) 28.19 (1.23) 26.75 (0.97)	Nasopharyngeal CT synthesis
Reaungamornrat et al. (2022)	CHUM&MGH (5461 slices)	MRI, CT	$256 \times 256 \times$ m	RaSGAN	Adv,reconst,feature	No	SSIM PSNR	0.74,0.83 26.09,25.04	Pelvis from MRI to CT synthesis

Table 4 (continued)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Li et al. (2022c)	Madic, BraTS 2020 IXI(40 subjects)	PET, CT images	256 × 256	TCGAN	Adv,GDL, pixel	No	SSIM PSNR VIF FID	0.966, 0.930, 0.867 45.66, 1.29, 27.37 0.836, 0.320, 0.342 69.94, 26.11, 55.23	Brain CT synthesis from PET

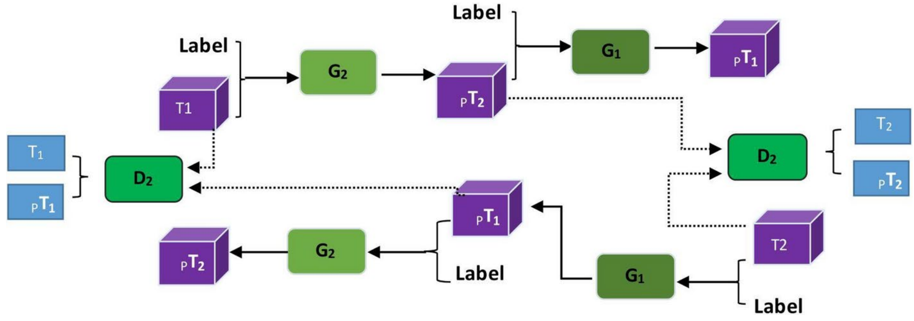

Fig. 7 Proposed framework by Xu et al. (2022d) (Synthesis)

4.1.2.3 X-ray image synthesis Due to X-ray images pose a risk of radiation exposure to patients, there is a limited availability of such data. To overcome this challenge, researchers often turn to GANs to artificially create additional data samples based on a given distribution. This helps to mitigate the scarcity of X-ray data. Jiang et al. (2022) proposed a generative attention network (UXGAN) for synthesizing ultrasound images of adolescent idiopathic scoliosis into X-ray images. A cycle-consistent network was used, and end-to-end training was performed. Attention modules were added and different residual blocks were designed, and successful results were achieved. However, GAN is a computationally intensive model that requires powerful hardware to run. Fernandes and Yen (2021) introduced a GAN pruning method that combines evolutionary strategy (ES) and multi-criteria decision making (MCDM). This approach achieves a good balance between computational complexity and synthesis performance without relying on weighting parameters. The experimental results demonstrated that the pruned GAN model successfully generated chest X-rays with comparable quality to the original model while reducing floating-point operations by 70%.

4.1.2.4 PET image synthesis PET, a type of medical imaging, is commonly used to diagnose and stage cancer. Additionally, combining different types of medical images like MR and PET can offer detailed information about the structure and function of the human body. However, in clinical practice, standard doses of radioactive tracers need to be injected into the body to obtain PET images, which inevitably cause radiation to the body and put patients at risk. Therefore, the medical imaging analysis community has been committed to synthesizing PET images.To avoid the radiation exposure caused by standard doses and reduce the noise and artifacts in synthesized PET images due to dose reduction, Luo et al. (2022) introduced a new method called Adaptive-regularized GAN (AR-GAN) with spectral constraint to produce high-quality PET images while minimizing radiation exposure. The model utilized spectral regularization to ensure the similarity between the generated image and the real image in terms of frequency content. This approach helps to retain frequency details and enhance the overall quality of the synthesized PET images. However, it is worth noting that many current applications using generative networks to synthesize PET images tend to overlook edge information. Amirkolaee et al. (2022) Proposed a new type of GAN that can create PET images from CT images without any intermediate steps. The model incorporates

Table 5 Medical image synthesis using conditional GANs–(MRI)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Zhu et al. (2021)	OASIS-brains (416 subjects) BraTS (285 patients)	MR images	$176 \times 208 \times 176$ 144×180 180×170 $240 \times 240 \times 155$	WGAN CNN	Adv L_1 per	Yes	PSNR FID	36.46, 39.85 34.28, 46.74	Brain MRI synthesis
Yurt et al. (2021)	IXI/ISLES (106 subjects)	MRI (T1, T2, PD-weighted, FLAIR)	256×256 256×256	mustGAN	Adv pixel-wise	Yes	PSNR SSIM	29.45 ± 1.19 0.947 ± 0.012; 35.89 ± 1.20 0.977 ± 0.005; 34.40 ± 0.97 0.974 ± 0.005	Brain MRI synthesis
Conte et al. (2021)	BRATS (train = 20925, validation = 5115, test = 6510 images)	(T1, T2, FLAIR) MRI	256×256	Pix2pix	Adv L_1 lad entropy	Yes	MSE(T1) SSI(T1) MSE(F) SSI(F)	0.005 — 0.013 0.004 — 0.103 0.82 — 0.92 0.76 — 0.92	Brain MRI synthesis
Luo et al. (2021)	BRATS2015 (220 subjects)	(T1, T2, FLAIR) MRI	n/a	EP_IMF GAN	L_1, L_2, Adv	No	PSNR SSIM NRMSE	23.563 (3.431) 0.930 (0.028) 0.287 (0.178); 26.063 (2.925) 0.947 (0.026) 0.294 (0.113)	Brain MRI synthesis
Xia et al. (2022)	UKBB(4102 SAX image) ADSB (200 subjects)	MRI (Cardiac)	256×256	cGAN, VGG-16, ConvLSTM	Adv bce LS L_1 per	No	n/a	Helps structural details	Cardiac MRI synthesis
Xu et al. (2022)	BraTS challenge (74 subjects) SPLP (private) (3360 images)	(T1, T2) MRI	70×70	Bi-MGAN	Adv Cyc per Handcraft feature	Yes	SSIM FSIM MSIM	57.1%, 9.1% 47.1%, 3.6% 50.0%, 9.6%	Brain MRI synthesis
Zhan et al. (2022)	BRATS2015 (20 volunteers) IXI (319 normal subjects)	(T1, T2, FLAIR) MRI	256×256	D2FE-GAN	Adv, PTMG, DMFE	No	PSNR SSIM NRMSE	26.655 (2.748) 0.950 (0.025) 0.278 (0.115) et al	Brain MRI synthesis

Table 5 (continued)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Amirkolaee et al. (2022)	Non-public (PET/CT/MR 69 subjects)	CT, MR, PET images	256×256	GAN DenseNet Edge detector network	Adv, L_2, CC, Gradient, Edge	No	RMSE MAE PSNR DSC NCC SSIM LPIPS FID	28.28 10.87 31.30 0.90 0.92 0.85 24.80 0.042	Brain MRI synthesis

an edge detection network to enhance the clarity of tissue edges in the synthesized PET images.

4.1.2.5 Ultrasound image synthesis Ultrasound (US) imaging is commonly used for examining body structures in clinical diagnosis. Deep learning algorithms for US image analysis usually need a substantial amount of data. However, acquiring and labeling a large volume of US images is challenging. As a result, Liang et al. (2022) developed an image synthesis method using GAN to create lifelike ultrasound (US) images. The model incorporates auxiliary sketch guidance to improve the fine details of the generated images. To produce high-resolution US images, a progressive training approach is employed to generate higher quality images step by step starting from lower resolution ones. The experimental results demonstrate the effectiveness of this method in synthesizing realistic US images. Furthermore, in order to tackle the problem of biased databases from various medical centers, Huang et al. (2022d) introduced an improved CycleGAN approach that enhances stability. This method effectively retains the fine details in the generated ultrasound images. The researchers performed experiments using clinical datasets of thyroid and carotid artery images. The outcomes demonstrate that the proposed model exhibits remarkable capabilities in transforming ultrasound images across different domains. It successfully mitigates the biases inherent in uniformly distributed datasets and exhibits improved stability during training.

4.1.2.6 Retinal image synthesis Due to the scarcity and class imbalance of retinal image data in the medical field, small training datasets severely affect the performance of deep learning network models, resulting in suboptimal realism and accuracy of image synthesis techniques. The Reference Zhang et al. (2023) introduces a network called MinimalGAN, which is designed to generate various medical images with accurate annotations, even when the training data is limited. The goal of MinimalGAN is to achieve high-quality image-to-image synthesis while using minimal training data. Unlike the minimal training data method proposed by Zhang et al. (2023), Guo et al. (2022) proposes a retinal image generation model called RetiGAN, which is based on GAN. To preserve the semantic information of the original image more effectively, RetiGAN incorporates the VGG network and utilizes content loss as a guide. This allows for the extraction of high-level semantic details from both the original and generated images. The experimental results demonstrate that the generated retinal images exhibit strong resemblance to the originals, exhibiting improved visibility and sharper edges. In terms of structural similarity and high resolution, RetiGAN surpasses other models for retinal image generation.

Moreover, age-related macular degeneration (AMD) is an eye condition that can result in vision loss. Employing advanced machine learning techniques to examine and categorize images of the retina aids tracking of AMD development.The reference Pham et al. (2022) presents a deep learning model that combines GAN with a drusen mask to preserve pathological information. This model, called Multimodal GAN (MuMo-GAN), synthesizes retinal images to predict the progression of AMD (Age-related Macular Degeneration) based on images taken at different times. The inclusion of the drusen mask helps in understanding the development of AMD. Through qualitative and quantitative experiments, it has been demonstrated that this model outperforms other studies in effectively monitoring the progression of AMD (see Table 6).

4.1.2.7 H&E images As computer technology advances quickly, GAN models have made remarkable strides in generating and transforming images. However, based on He et al. (2022b), when transforming microscope-scanned images, there is often a misunderstanding of the image information, leading to incorrect identification of content categories. To address this issue, researchers developed a new Pix2pix GAN model. This model utilizes UNet+ as the generator, which enhances feature extraction and gradient richness. Additionally, a partial regularization strategy is employed to train a portion of the generator network, allowing the model to accurately identify content categories during CARS-H&E image synthesis. Experimental results demonstrate a significant improvement in the quality of synthesized histopathology images generated from unlabeled CARS-H&E images. It is important to note that this method relies on paired training data, which can be challenging to acquire. Tsai et al. (2022) Uses CycleGAN to overcome the requirement for paired training data. They utilize unpaired OCT and H&E datasets to transform OCT images into H&E-like images with staining. Automated synthesis of histological images has numerous potential applications in the field of pathology. However, generating and modeling large, high-resolution images is a challenging task due to limitations in memory and computation resources. The authors in Deshpande et al. (2022) introduced a new approach called SAFRON (Seamless Across Field-Resolution Overlap Network) for generating realistic and high-resolution tissue images based on input tissue composition masks. The proposed framework is more efficient compared to current methods, as it requires less memory for training and performs faster in synthesizing large high-resolution images.

4.1.2.8 Blood vessels In the medical field, intravenous contrast agents are commonly used in CT imaging to assist in visualizing intravascular lesions, this greatly helps in diagnosing and treating patients. However, using contrast agents requires a significant amount of data. To address the data scarcity problem, Chandrashekar et al. (2023) proposed a DL pipeline based on GAN to generate CTA(Computed Tomography Angiography), greatly reducing radiation damage to patients as no contrast agent injection is needed. As generative networks continue to advance, they offer an alternative to CT scans in the form of MR angiography (MRA), which has become a valuable technique for visualizing blood vessels. Researchers Tang et al. (2021) created a type of artificial intelligence system called GANDA (Generative Adversarial Network for Distribution Analysis) to study and generate images of quantum dot (QD) distribution within tumors following intravenous injection. The deep generative model was automatically trained by decomposing tumor vasculature and nuclear blocks from full slide images of T1-weighted MR scans of breast cancer sections. This model allows to investigate the impact of various factors on the distribution of nanoparticles (NPs) within specific tumors, leading to improved molecular imaging and personalized treatment in the field of nanomedicine. In addition, You et al. (2022c) developed a deep learning model based on cycleGAN to generate synthetic TOF-MRA using PETRA for treating intracranial aneurysm patients. Moreover, optical-resolution photoacoustic microscopy (OR-PAM) is a powerful imaging technique with exceptional spatial resolution, gaining significant popular-

Table 6 Medical image synthesis using conditional GANs—(X-ray, PET, US, Retinal)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Fernandes and Yen (2021)	Non-public (5856 images) ISIC2016 (1279 images)	X-ray, skin lesion images	512×512	WGAN	Adv	No	Floating-Point Operations	70% fewer	Chest X-ray synthesis
Jiang et al. (2022)	Non-public (train 160 subjects, test 42 pairs images)	US to X-ray (AIS)	256×256	UXGAN	Adv cyc lsgan cam	Yes	NMSE PSNR SSIM	0.104 ± 0.059 29.74 ± 3.07 0.874 ± 0.015	AIS X-ray synthesis
Luo et al. (2022)	BrainWeb (20 subjects)	2D slice PET image	128×128	AR-GAN	AdvL_1 ce	No	PSNR SSIM	28.396, 28.106 0.896, 0.891	Brain PET synthesis
Liang et al. (2022)	Non-public, COVID-19 (6054 images) hip joint,Ovary (4492 images)	US image	256×256 512×512	GAN, Resnet, Patch-GAN	AdvL_1 feature	Yes	FID KID SSIM LPIPS	60.75,67.68, 61.48 13.36, 1.45, 5.36 0.7450, 0.6853, 0.5035 0.1693, 0.4401, 0.8177 et al	Lung, ovary hip joint US synthesis
Huang et al. (2022d)	Non-public (1656 unpaired, 78 pairs images)	US image	400×400	Stability-enhanced Cycle-GAN	L_1 least-square Adv	No	MAE sSSIM PSNR	12.20 ± 1.38 0.45 ± 0.06 19.47 ± 1.24	Thyroid Carotid US synthesis
Zhang et al. (2023)	DRIVE (40 examples) CVC-ClinicDB (912 images) CVC-ColonDB, ETIS-LaribPolyp-DB, ISIC 2018 (2694 images)	Retinal fundus images	640×640 384×384	Minimal-GAN	Adv fm per pixel	No	n/a	n/a	Synthesis Retinal fundus images
Guo et al. (2022)	DRIVE (40 images) MES-SIDOR (1200 images)	Retinal images	768×584 512×512	RetiGAN U-Net, VGG	Adv, content	No	SSIM FID KID	0.8365 46.74 0.0486	Synthesis Retinal fundus images

Table 6 (continued)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Pham et al. (2022)	Non-public (8196 images)	Fundus images (AMD)	224×224	MuMo-GAN	Adv, reconst	Yes	n/a	Help improved diagnosis performance	Synthesis Retinal fundus images

ity in recent times. However, due to the strong scattering of light in biological tissues, this application is limited to shallow depths. Cheng et al. (2022) Proposes the use of DL-enabled image transformation to achieve deep-penetration OR-PAM performance for blurry vascular images obtained using AR-PAM (acoustic-resolution photoacoustic microscopy).

4.1.2.9 Other Currently, deep learning models have demonstrated outstanding performance in image processing tasks. However, creating these models in the medical field often faces common issues: patient privacy protection and data class imbalance (Torfi et al. 2022; Zhang et al. 2022f). To overcome these problems, many researchers typically use generative networks to synthesize the desired target images and then use the synthesized images to further assist in improving the model's performance. Gan and Wang (2022) has demonstrated that in recent times, advanced machine learning techniques have been extensively utilized to analyze esophageal endoscopic OCT (optical coherence tomography) images, yielding favorable outcomes. However, it is important to note that this approach necessitates a substantial volume of training data. Samples generated using traditional data augmentation techniques sometimes deviate significantly from reality, resulting in unsatisfactory training models. As a result, the researchers have come up with a straightforward and efficient approach to produce top-notch esophageal OCT samples by combining GAN and VAE (variational autoencoder) techniques. The experimental outcomes have validated that this method yields superior image quality when generating esophageal OCT images and significantly enhances the performance of the deep learning model. According to Huang et al. (2022e), arterial spin labeling (ASL) images have become more popular in research for diagnosing dementia. However, this valuable type of imaging is not commonly included in many established dementia datasets that are based on images. Therefore, the authors have introduced a new integration method based on locally constrained GAN to synthesize ASL images to supplement these datasets critical modalities and improve dementia diagnosis accuracy. Experimental results have shown that synthesizing ASL images with the help of the novel GAN integration method can significantly improve dementia diagnosis accuracy. Additionally, Qadir et al. (2022) has also developed a DL-based framework for generating polyp images. The framework utilizes a straightforward conditional GAN design to generate synthetic polyp images that closely resemble actual polyps. The outcomes have demonstrated that incorporating these generated polyp images into the training dataset can significantly improve the performance of models (see Table 7).

4.2 Segmentation

Currently, in the field of medicine, the segmentation of medical images of patient tissues and organs. You et al. (2022d, e, 2023c) is a prerequisite for a lot of applications such as disease detection, analysis, treatment, and classification. The goal is to make the anatomical or pathological changes in the images more clear, which can significantly enhance the accuracy and efficiency of diagnosis. Early methods of medical image segmentation were typically based on region, edge, classification, and traditional machine learning algorithms. Although these methods achieved good results to some extent, performance is still affected by weak edges, pseudo-edges, and noise in medical image segmentation tasks. Additionally, selecting an appropriate threshold is also very difficult. With the rapid development of DL technology, GANs have gradually been applied, which has largely addressed the aforementioned issues. In the following section, we will discuss recent studies that explore the use of GAN and its related versions for medical image segmentation applications, including segmentation tasks for brain, eye structures, thyroid glands, breasts and breast tumors, lungs, hearts, abdominal organs, lumbar vertebrae, and cells. This helps researchers improve the segmentation accuracy of medical images and provide reliable diagnostic and treatment basis for radiation therapy doctors, thereby achieving the goal of shortening diagnosis time and precise treatment.

4.2.1 Brain

Brain tissue segmentation in medical image processing can assist doctors in making accurate diagnoses and treatments for brain diseases. It has important practical value in the field of clinical brain medicine. Compared to CT images, MRI is the main imaging tool used by clinical doctors to study brain tissue structure. How to segment MRI brain tissue is not only a focus and difficulty in MRI brain tissue image research, but also in clinical diagnosis and treatment. Previous research often utilized single-step models for segmenting brain images, which could result in substantial loss of important information. In their study, Khaled et al. (2022) introduced a multi-stage GAN model to address the limitations of single-stage models in brain image segmentation. The model goes through multiple stages to improve the segmentation process. In the first stage, it generates a basic outline of the background and brain tissue. Then, it refines the contours of cerebrospinal fluid (CSF), gray matter (GM), and white matter (WM) with more precision. Finally, the rough and refined contours are combined to achieve more accurate segmentation results. The above proposed model has shown good segmentation performance in practice, but can only be trained using labeled data. In addition, the labeled image data is severely limited. Therefore, using 2D GAN to synthesize labeled image data can solve the problem well. Platscher et al. (2022) Demonstrated in MRI studies of stroke injury brain volume that the use of deep learning-based enhancement techniques can improve lesion segmentation. However, 2D GANs are only suitable for 2D images and cannot be applied to 3D medical images. Çelik and Talu (2022) Introduced a new GAN-based segmentation architecture, Vol2SegGAN, which can

Table 7 Medical image synthesis using conditional GANs—(H&E, blood vessels, other)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Tang et al. (2021)	Non-public (27775 patches)	Tumor vessels, cell nuclei, intratumoral QDs distribution images	512×512	GAN,FCN	Adv, pixel-wise	No	MSE	1.871	Breast images synthesis
He et al. (2022b)	Non-public (200 raw image pairs)	DCARS-H&E corresponding mask image	256×256	Pix2pix, GAN	Adv, ce, MSE	No	MSE SSIM	0.0074 ± 0.006 0.8810 ± 0.072	Thyroid DCARS-H&E synthesis
Tsai et al. (2022)	Non-public OCT dataset, H&E stained dataset	OCT,H&E image	576×200	CycleGAN, VGG16	Adv, cyc, FM	No	IOU PCC	82% 84%	Skin H&E synthesis
Deshpande et al. (2022)	CRAG (128 images), Digestpath challenge	Tissue component mask, histology image	1024×1024	SAFRON (GAN,U-Net, PatchGAN)	Adv, rec	No	FID	10.75 ± 0.13 12.99 ± 0.13	Colorectal histology image synthesis
You et al. (2022c)	Non-public (3298 subjects)	PETRA-MRA To TOF-MRA Synthesis	512×512	cycleGAN	Adv, cyc, id, class, activation map	No	PSNR SSIM	17.51dB 0.71 ± 0.02	Intracranial Aneurysm PETRA-MRA To TOF-MRA synthesis
Gan and Wang (2022)	Non-public (11000 images)	OCT image (esophageal)	256×256	AL-VAE (GAN, VAE)	Adv, prior, like	No	WD MMD	0.1234 0.2465	Esophageal image synthesis
Huang et al. (2022e)	Non-public (355 patients) ADNI-1	ASL,MRI images	64×64×21	GAN, FP-GAN	Adv	No	n/a	Help improved diagnosis performance	Arterial Spin Labeling image synthesis
Qadir et al. (2022)	CVC-ClinicDB (612 images) ETIS-LARIB (196 images) CVC-ClinicVideoDB (18 SD)	Polyp/masks, negative images	388×284 384×288 1255×966	cGAN, CNN	Adv, L_1, bce, rec, Jaccard, WGAN-GP	No	n/a	n/a	Polyp negative images synthesis

be used for 3D-MRI brain tissue segmentation. First, the model proposes preprocessing of brain MRI, including extraction of brain regions, cropping/sampling. Then, a GAN framework with ACFP and PAM modules can be used for 3D-MRI brain tissue segmentation. Meanwhile, in a different study, Subramaniam et al. (2022) developed a 3D WGAN model that utilized mixed-precision algorithms to generate lifelike 3D TOF-MRA patches for the purpose of segmenting brain vessels. Segmenting brain vessel images is beneficial for diagnosing and treating brain vascular diseases. However, due to the intricate and small nature of the brain vessel structure, segmentation becomes challenging. Hence, Chen et al. (2022d) introduced a deep learning approach based on GANs to convert TOF-MRA data into 3D representations of brain vessels. The neural network employs Vnet to extract and combine features at various scales, enhancing the ability to express features effectively. Subsequently, the network evaluates the segmentation results of brain vessel images based on criteria such as the smallest vessel diameter and the number of isolated regions, ensuring accurate vessel recognition and spatial consistency. The model uses TOF-MRA data for brain vessel segmentation training and obtains an average dice similarity coefficient of 89.89%, which is a good and accurate medical image segmentation result.Accurate segmentation of medical images is extremely important for diagnosing cancer and other brain diseases. Currently, doctors manually annotate and segment brain tumor tissue based on their personal experience, which is a time-consuming process and can potentially result in misdiagnosis. Therefore, Güven and Talu (2023) proposed a new GAN segmentation architecture, SSimDCL, for automatic brain MRI image segmentation. SSimDCL achieved satisfactory results in the proposed research (Fig. 8).

4.2.2 Eye

Morphological features of blood vessels in the retina are crucial for diagnosing eye diseases at an early stage. In both clinical and research settings, accurately identifying different tissue layers in optical coherence tomography (OCT) images of the retina and choroid is essential. However, a significant drawback of using machine learning (ML) algorithms for segmentation is that the performance of the model heavily relies on the quantity and diversity of data available for training. Kugelman et al. (2021) It is suggested to use GAN here to increase data for patch-based OCT choroid and retina boundary segmentation methods. A series of experiments were carried out to gain insights and improve the performance through optimization. The findings indicated that it is possible to create patches that look very similar to real ones, and in the ideal scenario, the performance of segmentation using only synthetic data is nearly as good as models trained on real data. Beji et al. (2023) A new GAN-based segmentation network Seg2GAN was also developed, which uses enhanced retinal and coronary artery vessels as well as knee joint cartilage datasets for training. The experiment achieved good segmentation results. However, in low-contrast backgrounds and lesion areas, segmentation performance is highly affected by vessels. In addition, a notable study addresses the challenge of domain adaptation for optic disc and cup segmentation in fundus images (Kadambi et al. 2020). This research demonstrates the effectiveness of using Wasserstein Generative Adversarial Networks (WGAN) to reduce domain gaps, thereby significantly improving segmentation performance. The WGAN approach focuses on better feature extraction and adaptation, which has led to enhanced accuracy in glaucoma diagno-

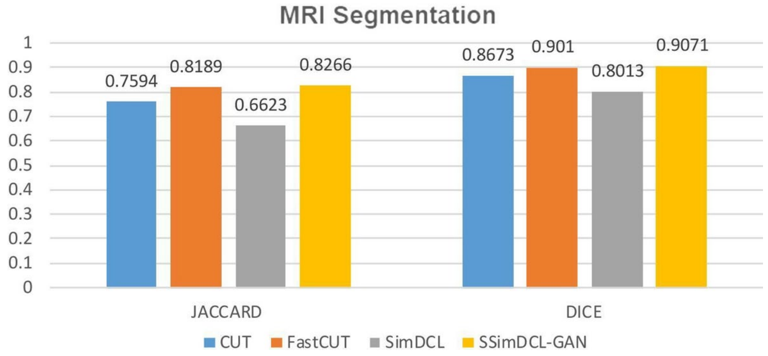

Fig. 8 Güven and Talu (2023) proposed a model combining SSIMDCL with GAN for brain MRI segmentation. The experiment used JACCARD and DICE as evaluation metrics. By comparing with non-GAN models CUT, FastCUT, and SimDCL, it was observed that the SSIMDCL GAN model achieved excellent results

sis when compared to traditional methods. Such advancements underscore the potential of GAN-based models in bridging domain disparities and improving clinical outcomes.

4.2.3 Thyroid

The precise segmentation of nodules in thyroid ultrasound (US) images is of great clinical significance in aiding doctors in thyroid diagnosis.In addition to immediate diagnosis, automated tools can also track the possibility of malignant tumors over time. Recently, DL-based image segmentation techniques typically use convolutional neural networks (CNNs) for training. However, CNNs rely on a substantial amount of annotated images for effective training. However, it is extremely challenging and expensive to acquire a substantial amount of fully labeled images, making it difficult for the model to obtain sufficient training data. Hence, Liu et al. (2021c) introduced the concept of uncertainty in the fine GAN (U2F-GAN). This approach utilizes a weakly supervised framework to accurately segment nodules in thyroid ultrasound (US) images. The researchers employed a technique called superpixel processing to capture the basic structural characteristics of the images during model training. They also incorporated a module for comparing similarities and a loss function that constrains the distribution to improve the overall accuracy of segmentation. The segmentation results of the U2F-GAN approach are on par with fully supervised methods, outperforming other weakly supervised methods that have been used previously. Furthermore, a recent study by Kunapinun et al. (2023) presented a novel technique called StableSeg GAN, designed for accurately identifying thyroid nodules in ultrasound images. The researchers proposed a fusion of traditional supervised semantic segmentation and unsupervised learning techniques using GANs. To ensure stable training of the GAN model and prevent mode collapse, they introduced the concept of closed-loop control for the discriminator loss output gain. Additionally, by combining supervised and unsupervised learning approaches, the model promoted both high-level consistency and low-level accuracy in the segmentation results. Experimental results showed that StableSeg GAN has good flexibility in thyroid nodule segmentation tasks.

4.2.4 Breast

Using AI technology to accurately locate and separate breast tumors in medical images of the breast can assist doctors in diagnosing and treating breast cancer, ultimately reducing mortality rates. While deep learning-based segmentation networks have made significant advancements and yielded promising results, the limited availability of clinical datasets hinders further progress in developing breast medical image segmentation algorithms. Therefore, Baccouche et al. (2021) proposed using CycleGAN models to synthesize breast X-ray images for training and developing Connected-UNets automatic segmentation architecture, which overcomes the problem of the lack of breast X-ray images in clinical practice in breast medical imaging datasets. The authors experimented with the segmentation model on two publicly available datasets and an enhanced dataset, and the results showed that the enhanced dataset helped improve segmentation quality. In radiation therapy planning, ultrasound imaging is also an important method for diagnosing breast cancer. However, marking ultrasound images is expensive and there is a limited amount of data in each dataset, which impacts the effectiveness of segmentation algorithms. Therefore, Zhai et al. (2022) introduced a new type of semi-supervised GAN called ASSGAN. This model utilizes two generators and one discriminator to learn from each other in an adversarial manner (see Fig. 9). ASSGAN is capable of generating accurate segmentation prediction masks even without the need for labeled data. Furthermore, the proposed method demonstrates strong performance even when only a limited number of labeled images are available. It's important to consider that when using breast ultrasound images as input data, there may be a blurring effect on the edges of some tumor lesions in the images. This blurring makes it challenging to accurately segment the breast lesions. However, a new segmentation framework proposed in Wang et al. (2022a) for tumor lesion segmentation in breast ultrasound images can effectively solve the aforementioned problem. The authors utilized a special type of neural network called cGAN as the primary framework of their model. They incorporated a deformable U-Net in the generator, which allowed for precise pixel-level segmentation of the breast US images. To enhance the performance of the segmentation network, they introduced a Markov discriminator that provided discrimination loss during training. It is worth mentioning that MRI is also utilized for breast tumor segmentation in this study. For instance, in their study Haq et al. (2022), researchers developed a breast tumor segmentation model called BTS-GAN that automatically identifies tumors in MRI scans. To improve the accuracy of localization, they introduced skip connections between the encoder and decoder as part of the generator. Additionally, they employed a parallel dilated convolution (PDC) module to preserve features and extract information about high-quality edges and internal textures. The experiments demonstrated that BTS-GAN outperformed other methods, achieving an average Intersection over Union (IoU) score of 77% and a Dice score of 85% in breast tumor segmentation.

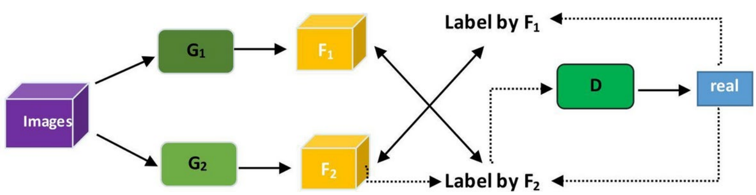

Fig. 9 Proposed framework by Zhai et al. (2022) (Segmentation)

4.2.5 Lung

Accurate identification of the lungs in CT scans is essential for detecting early-stage lung cancer, as it directly impacts the patient's chances of survival. Tan et al. (2021) to find an automatic method for segmenting lungs in CT images, researchers developed a new segmentation approach using a deep learning GAN (LGAN). However, existing methods struggle to accurately segment lungs in CT scans when there are dense abnormalities present. Therefore, Pawar and Talbar (2021) proposed the LungSeg-Net method based on generative adversarial networks for accurate segmentation of the lung region from the surrounding chest area. The LungSeg-Net model takes a lung CT slice as input and processes it through an encoder to create a set of feature maps. These feature maps are then passed through the multi-scale dense feature extraction (MSDFE) module to extract features at different scales. The decoder then uses these multi-scale features to generate a lung segmentation map. To evaluate the LungSeg-Net model, it was tested on the ILD dataset, which is widely used in the field. The experimental results confirmed that the LungSeg-Net model performs consistently well even when dealing with dense abnormalities in lung CT scans.

Currently, segmenting lung CT scans can help detect lung diseases, and segmenting chest X-ray images Dai et al. (2017) Mansouri et al. (2022) can aid in the early detection of lung cancer. In Lian et al. (2022a), a model that combines segmentation and classification tasks was introduced. They used the DRD U-Net model to identify and separate abnormalities in lung CT scans and X-ray images. While convolutional neural networks have shown impressive results in medical image segmentation, limited data availability and imbalanced classes can cause problems like overfitting and subpar performance. In order to address these issues, researchers in Tyagi and Talbar (2022) created a special type of computer network called CSE-GAN, which can accurately identify and outline lung nodules in 3D images. This network incorporates a technique called concurrent squeezing and excitation blocks, allowing it to better understand the patterns in the data and produce more precise segmentation results. This improvement in accuracy is achieved by training the network to learn the characteristics of lung nodules and apply this knowledge during the segmentation process. To address the problem of overfitting, the researchers used a training approach based on dividing the data into patches. They tested their method on two datasets: LUNA16 and a local dataset. The results showed that the dice coefficients for the LUNA test set and the local dataset were 80.74% and 76.36% respectively, indicating a significant improvement in model performance. Similarly, the sensitivities for these datasets were 85.46% and 82.56% respectively, further confirming the enhanced performance of the model.

4.2.6 Cardiology

Accurately identifying and delineating different parts of the heart is essential for diagnosing and planning treatment for heart diseases. However, the lack of labeled cardiac magnetic resonance imaging (CMRI) data and the difficulty in dealing with variations in size, shape, and deformation of the left ventricle (LV) pose challenges for traditional convolutional neural networks (CNNs) to automatically segment the LV from CMRI images. In order to improve the segmentation of the left ventricle (LV) in cardiovascular imaging, a new method for automatically segmenting the LV is proposed in Wu et al. (2021a). This method utilizes adversarial learning, where a generative adversarial network (GAN) is trained on both labeled and unlabeled CMRI data. By incorporating both ground truth and unlabeled samples during training, this method not only achieves faster convergence but also improves the performance of LV segmentation. Compared to the study mentioned in Wu et al. (2021a), Cui et al. (2021) proposes a new framework called GBCUDA (GAN-based bidirectional cross-modal unsupervised domain adaptation) for segmenting heart images. The main objective is to solve the issue of decreased segmentation performance when the network adapts to a target domain without ground truth labels. GBCUDA utilizes adversarial learning to extract image features and gradually improve the domain invariance of feature extraction. Additionally, the authors introduce a self-attention mechanism into the GAN network, which allows for generating detailed information based on cues from all feature positions. They also implement spectral normalization to stabilize GAN training and introduce knowledge distillation loss to better handle advanced feature maps for cross-modal cardiac image segmentation tasks. In cardiovascular imaging segmentation, segmenting heart CT images Le et al. (2021) and segmenting heart MRI scans are the most common ways of diagnosing and analyzing heart diseases. Determining if the heart is normal can be done by analyzing the size and shape of its ventricles in segmented heart MRI images. However, the ventricle's boundaries are unclear and blurry due to the lack of contrast with the background region. This makes it difficult to accurately segment the ventricle. Thus, Zhang et al. (2022a) proposes the transfer learning and multiscale discriminative generative adversarial network (TLMDB GAN) method for segmenting MRI ventricular images. The authors in Li et al. (2021) suggest a network for segmenting multiple targets in cardiac MRI data using a semi-supervised learning approach. They use U-Net as an encoder and cGAN as a decoder. To align the heart images, they apply an affine transformation, and a limited amount of annotated MRI data is used for pre-training the GAN. The proposed semi-supervised learning network exhibits similar generalization ability to supervised learning-based networks. Additionally, the unsupervised domain adaptation framework based on adversarial networks proposed by Dong et al. (2018) and the SimCVD (Simple contrastive voxel-wise representation distillation) framework proposed by You et al. (2022a) both demonstrate the promising prospects of semi-supervised learning in medical image segmentation You et al. (2022f, 2023a, b, 2024).

4.2.7 Abdomen

Automated computerized cutting out of fat tissue (AT) from CT scans can be utilized to help with medical diagnosis. However, in deep learning segmentation networks, CT images require professionally accurate annotations to effectively improve segmentation performance. Hence, Shen et al. (2022) introduced a semi-supervised segmentation network called URO-GAN, which utilizes adversarial learning to segment abdominal CT images and focuses specifically on the segmentation of AT. The authors introduced an unreliable region optimization mechanism in the model and effectively improved segmentation quality by using annotated and unannotated images for training. Currently, the small size of annotated datasets required for training deep learning segmentation networks affects network segmentation performance and hinders the expansion of deep learning. Song et al. (2022) suggested using transfer learning techniques to enhance the accuracy and generalization capability of a kidney ultrasound image segmentation model when there is not enough training data. The Li et al. (2022a) model proposed using a generative component to synthesize high-resolution, large-scale prostate tissue pathology images, while also performing semantic segmentation of tissue pathology images. Experiments have shown that synthetic images and their masks can improve the model's segmentation performance.

Accurately identifying and separating the liver area from abdominal CT images Kong et al. (2022a) is crucial for the computer-assisted diagnosis and treatment of liver cancer using medical imaging. However, because of the nature of liver tissue and the limitations of CT image processing, the liver region in CT images has uneven shades of gray and unclear boundaries with neighboring organs, posing significant challenges for liver segmentation (You et al. 2020a, f). In the referenced paper He et al. (2022a), the authors aimed to improve the accuracy of segmenting 3D images using the 3D U-Net network. They proposed an enhanced version of the 3D U-Net network and integrated it into a generative adversarial network (GAN) framework. This allowed them to develop a semi-supervised algorithm for optimizing 3D liver segmentation. After conducting experiments on the LiTS-2017 and KiTS19 datasets, He et al. (2022) researchers demonstrated that their new algorithm for liver segmentation, which uses both labeled and unlabeled data, significantly enhances the accuracy of liver segmentation. Liver CT images offer abundant details at a reasonable cost. On the other hand, MRI provides more comprehensive quantitative information about the liver compared to CT. In a recent study by Hong et al. (2022), they introduced a new method for improving the segmentation of liver MRI images across different imaging modalities. Their approach involves combining adversarial learning, self-learning, and the use of adversarial losses based on semantic perception and shape entropy perception. This allows the network to be trained in an unsupervised manner, without the need for labeled data. Currently, in the medical field, segmenting CT and MRI images You et al. (2022b) is crucial for various purposes such as diagnosis, prognosis, surgical planning, treatment evaluation, and more (Jafari et al. 2022). While CNNs have been widely used for image segmentation, there is a recent trend towards transformer-based architectures. Transformers have the ability to capture long-range dependencies in signals through attention mechanisms, making them increasingly popular in the field. In Demir et al. (2022), a new technique for segmenting the liver was introduced. This method combines GANs with Transformers. The Transformers utilize a self-attention mechanism to gather important information from different parts of the image, resulting in better segmentation performance compared to traditional methods.

4.2.8 Spine

Segmenting lumbar spine images helps diagnose spinal diseases. However, accurately identifying multiple spinal structures is challenging due to the complicated structure of axial MRI images, unclear boundaries between different structures, and the risk of overfitting. To address these issues, Gong et al. (2022) proposed using the ResAttenGAN network framework for accurate segmentation of the dura mater, neural foramina, lamina, and intervertebral discs. First, the Full Feature Fusion (FFF) module fuses all hierarchical structures and generates distinct feature representations. Second, the Refined Region Attention (RRA) module accurately locates implicit boundaries and refines its per-pixel classification. Finally, the AL (Adaptive Layer) module helps improve the accuracy of the model by making sure that the spatial details are consistent without overfitting. When examining CT scans at the third lumbar vertebra (L3) level, it helps in measuring the amount of muscle and fat tissue in the body. However, manually selecting the L3 vertebra is time-consuming. Therefore, Zhang et al. (2022b) proposed a new method that automatically locates the L3 vertebra and accurately segments body tissues using a segmentation network. Introduction of adversarial structures and cross-stage attention (CSA) blocks improved tissue segmentation performance. Changes in paraspinal muscle cross-sectional area (CSA) and fat infiltration (FI) following posterior lumbar spine surgery (PLS) can profoundly affect muscle activity patterns and spinal stability. Automated segmentation of paraspinal muscles (multifidus and erector spinae) assists in disease treatment. However, due to the overlap of paraspinal muscle distribution with other anatomical structures, unclear fascia between multifidus (MF) and erector spinae (ES), and variable shape within and between patients, semantic segmentation of MF and ES faces challenges. Li et al. (2022d) proposed a deep learning model called LPM-GAN to automatically segment paraspinal muscles in MRI images, aiming to address the above issues. The generator component of the model extracts meaningful information from the images while preserving the anatomical structure of the paraspinal muscles, effectively dealing with their high variability and diversity. The discriminator component is trained to refine the predicted muscle segmentation by aligning it with the ground truth. Additionally, the model utilizes the Otsu method to calculate the cross-sectional area (CSA) and fat infiltration (FI) of the paraspinal muscles (see Tables 8, 9, 10, 11, 12, 13, 14, 15).

4.2.9 Cell images

A large annotated dataset can effectively improve the performance of deep learning methods in medical image tasks. Obtaining high-quality annotations in cell segmentation tasks is expensive and complex. Therefore, a new method called A-GAN was proposed by Liu et al. (2021a), which combines cGAN with an ACAM (active cell appearance model).By combining the outlines created by ACAM with the generated images from A-GAN, can generate high-quality annotated images. The experiments conducted on adaptive optics (AO) retinal images demonstrate that A-GAN can reliably produce realistic manually annotated images, thereby enhancing the accuracy of cell segmentation. The authors of Zhang et al. (2021) introduced a new technique for segmenting cell nuclei called MASG-GAN (multi-view attention superpixel-guided GAN), which combines multiple tasks into a single learning process. The model utilizes an innovative approach to superpixel segmentation using a bounded asymmetric Gaussian mixture model (BAGMM), resulting in high-quality superpixel prior probability maps with detailed semantics. Then, the generator network with integrated student and teacher branches is used for accurate segmentation and classification. Finally, the MVAM (multi-view attention module) is used to further improve the quality of cell nucleus segmentation. Currently, deep learning-based cell nucleus segmentation methods generally perform better than traditional methods, but may have issues with under-segmentation and over-segmentation. Therefore, Wang et al. (2022b) proposed a novel integrated network based on GAN and modified U-Net for cell nucleus segmentation. The SC-MT attention mechanism introduced in the model alleviates under-segmentation and over-segmentation problems. In traditional biomedical research, segmenting and counting cells are crucial and time-consuming steps in experiments. To address this issue, the researchers in Deng et al. (2022) developed a GAN-based model that is efficient, lightweight, and incorporates multi-scale feature fusion. This model focuses on two tasks: cell counting and segmentation. The proposed network is trained using generative adversarial networks. Multi-scale feature fusion technology is used for medical image segmentation.

Table 8 Methods that use GAN-based for segmentation (Brain)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Khaled et al. (2022)	MICCAI iSEG-2017 (23 images) MRBrainS-2013(20 images)	T1/T2 MRI	1.25 × 1.25*	GAN U-net	Adv label fake	No	CSF GM WM	95%, 93% 94%, 93% 92%, 88%	MRI segmentation
Platscher et al. (2022)	Non-public (449 positive, 85 DWI negative samples)	lesion masks	128 × 128 × 40	Pix2Pix, CycleGAN	Adv L_1 rec, id, ce	Yes	DSC	72.8%	Lesion segmentation
Çelik and Talu (2022)	IBSR18 (18 subjects) MR13(20 individuals) MR18 (7 subjects)	3D MRI	256 × 256 ×128 240 × 240× 48 240 ×240 ×48	Vol2SegGAN	Adv	Yes	Dice (CSF,GM) VS(CSF) HD (GM,WM)	0.739, 0.878 0.967 2.378, 2.105	MRI segmentation
Subramaniam et al. (2022)	PEGASUS (n = 72) 1000Plus (n = 65)	3D TOF-MRA	128 ×128 ×64	3D WGANs	Adv	Yes	DSC bAVD	0.841 0.508	3D TOF-MRA segmentation
Chen et al. (2022d)	FAH-WMU(80 cases) FAH-ZJU(20 cases)	TOF-MRA,3D cerebral vessels	560 ×560 ×140	SEVnet-GAN	Adv, Dice content, Least square	No	DCS	89.89%	TOF-MRA segmentation
Güven and Talu (2023)	Non-public (train = 1648, test = 366)	3D was converted to 2D MRI	256 ×256	SSimDCL (DCLGAN)	Adv, id, sim, L_1 PatchNCE	No	LPIPS PSNR FID KID JACCARD DICE	0.042 37.138 20.07 0.007 0.9880 0.9940	3D MRI segmentation

*Represents a multiplication factor of 1.25. This indicates that the third dimension (or depth) of the resolution is also 1.25

Compared to U-Net, our method achieved better results in segmenting units. Moreover, the proposed proposed approach enables counting without relying on individual points and eliminates the need for precise labeling of each cell's position during training. The experimental results demonstrate outstanding performance in both cell counting and segmentation (see Table 16).

4.2.10 Other

Annotated data in large quantities can help improve the performance of DL in medical image segmentation. However, obtaining and annotating such data is time-consuming and difficult. Therefore, Wang et al. (2022c) developed a new and improved model called AU-MultiGAN (asymmetric U-Net GAN) for medical image segmentation. The AU-MultiGAN utilizes multiple discriminators and an asymmetric U-Net architecture to generate multiple segmentation maps simultaneously. By incorporating the multi-discriminator module into the asymmetric U-Net structure, the model effectively captures and transfers valuable information from the input samples. The experimental results demonstrate the effectiveness of the AU-MultiGAN method in medical image segmentation, surpassing the performance of existing state-of-the-art approaches.

4.3 Classification

We reviewed many GAN-based frameworks for medical image data generation (Wang et al. 2022d) in Sect. 4.1. However, some researchers in the medical field have also applied GANs to classification tasks. In recent years, deep learning (DL) methods have been shown to be applicable to medical image classification tasks (Saleem et al. 2022). However, the lack of large annotated training data (Pang et al. 2021; Zhang et al. 2022c) and the problem of imbalanced training data (Xia et al. 2023; Chen et al. 2023) have seriously hindered its development prospects. To overcome these limitations, in addition to manually increasing the training data (Fig. 10), some research has also used GAN-based extended networks to synthesize the target data needed to assist convolutional neural networks (CNN) in image classification (Pang et al. 2021), or used a small amount of labeled data to train transfer learning models to classify large amounts of unlabeled image datasets (Wang et al. 2021b). Some research even directly embeds optimization algorithms into GAN frameworks to improve classification performance (Neelima et al. 2022; Chen et al. 2022a). In the following subsections, we will refer to some papers for a detailed description.

4.3.1 GANs+CNN combination models

Timely detection and treatment of diseases can effectively reduce patient mortality rates. The use of CNN classification algorithms for arrhythmia detection far outperforms traditional methods, but requires sufficient annotated image data, which is a time-consuming process. Pang et al. (2021) proposes the use of a GAN-based semi-supervised radiomics model to synthesize high-quality breast ultrasound images, which are then used to train a CNN for breast tumor classification. The experiment obtains more accurate results for breast tumor classification using the semi-supervised synthesized data. To address the problem of imbalanced labeled electrocardiogram (ECG) data in training, Ma et al. (2022) proposes

Table 9 Methods that use GAN-based for segmentation (Eye)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Kugelman et al. (2021)	MNIST (594 scans)	OCT images	1536 × 496	GAN	Adv, ce	No	Acc	87.56%	OCT segmentation
Zhou et al. (2021a)	DRIVE (40 images) STARE (20 images) CHASEDB1 (28 images) HRF (45 images)	Retinal fundus images	880 × 608	SEGAN, AM, MSFRB	Adv, reg, pixel classification	No	Se, Pr, AUC,F1	0.8294, 0.8397, 0.9830, 0.8345; 0.8812, 0.7952, 0.9863, 0.8359; 0.8435, 0.8013, 0.9872, 0.8218; 0.8310, 0.8115, 0.9693, 0.8211	Retinal fundus images segmentation
Beji et al. (2023)	DRIVE (25 images)	Retinal images	768 × 584	Seg-2GAN	Adv, L1, ce	No	Dice	0.878	Retinal images segmentation

Table 10 Methods that use GAN-based for segmentation (Thyroid)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Liu et al. (2021c)	Non-public (500 images)	Ultrasound image(US)	256 × 256	GAN,K-means	Adv, distributed	No	Dice HD	0.87 13.47	US segmentation
Kunapinun et al. (2023)	Non-public	US	256 × 256	StableSeg GAN	Adv, bce, dice, L_1	No	IoU	81.26%	US segmentation

using ECG-DCGAN to augment the ECG dataset. Then, a CNN model is used for automatic feature extraction and classification of ECG data, achieving 98.7% accuracy. Xia et al. (2023) Proposes using a Transformer and Convolutional GAN (TCGAN) method to enhance each type of ECG data and alleviate the data imbalance problem. In previous studies, using traditional data augmentation techniques alone was difficult to obtain high-quality images. Therefore, Barshooi and Amirkhani (2022) combines traditional data augmentation techniques with GAN to synthesize chest X-ray (CXR) images, which are then preprocessed with Gabor filters and used to train a CNN. The proposed method achieves 98.5% accuracy in a binary classification task. Hazra et al. (2022) Proposes the use of a GAN-based extension network, C-WGAN-GP, to enhance the number of microcell images in bone marrow

Table 11 Methods that use GAN-based for segmentation (Breast)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Baccouche et al. (2021)	CBIS-DDSM (1467 lesion) Inbreast (112 lesion) Non-public (638 lesion)	X-ray	256×256	CycleGAN, AUNet, ResU-Net	Adv, Dice, IoU, cycle	Yes	Dice IoU	89.52%, 95.28%, 95.88% 80.02%, 91.03%, 92.27%	X-ray segmentation
Zhai et al. (2022)	DBUI(647 cases) ADBUI(200 cases) SPDBUI (320 cases) SDBUI(1805 cases)	US	320×320	ASS-GAN	Adv, ce	No	IoU,Acc Dice	0.7683, 0.9760 0.8690; 0.6187, 0.9605 0.7644; 0.8852, 0.9508 0.9391; 0.7123, 0.9589 0.8319	US segmentation
Wang et al. (2022a)	Non-public (Train = 2916, validation = 128, test = 198 images)	US	256×256	cGAN, Markov, Deformed U-Net	Adv,L_1, L_2	No	DICE, F1,Acc SP,PC Recall Mean-IOU	89.7%, 89.7%, 98.1% 98.5% 86.3%, 94.7% 82.2%	US segmentation
Haq et al. (2022)	RIDER (500 images)	MRI	256×256	BTS-GAN	Adv, L1, ce	No	IoU, Dice	77%, 85%	MRI segmentation

aspirate smears, for the purpose of cell classification and counting to aid in the diagnosis of blood diseases. The enhanced data is then used as input for training the CNN classifier.The experiment achieves higher classification accuracy on the augmented dataset compared to the original data, with a classification accuracy of 96.98% (see Table 17).

4.3.2 GAN extension models

Currently, although classification networks based on Convolutional Neural Networks (CNNs) have achieved some decent classification results Hazra et al. (2022), the low-resolution of cell images severely affects the performance of CNN-based classification. Therefore, an end-to-end GAN-based synchronous super-resolution cervical cell classification algorithm was proposed in Lin et al. (2022). Ordinal regression using smooth L1 loss was used to further improve the model's classification performance. In some studies, deep convolutional networks are also applied to medical image processing tasks. However, similar to Xia et al. (2023), Chen et al. (2023), Pang et al. (2021), the lack of data and imbalance lead to a decrease in the network's diagnostic and classification performance. Unlike deep convolu-

Table 12 Methods that use GAN-based for segmentation (Lung)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Tan et al. (2021)	LIDC-IDRI (220 scans) QIN Lung (47 scans)	CT scans	512 × 512	LGAN	Adv EM BCE	No	(Mean Median) 3D Dice	0.985 0.9864	CT segmentation
Pawar and Talbar (2021)	ILD(4000 CT slices)	HRCT slices	512 × 512	LungSeg-Net (cGAN,Encoder /decoder, MSDFE)	Adv, L_1	No	Dice Jaccard	0.9899 0.98	HRCT slices segmentation
Lian et al. (2022a)	Non-public (4251 image)	X-ray, CT	n/a	WGAN,DNN, DRD U-Net	Adv, ce	No	Accuracy F-measure G-mean	73.84% 74.65% 74.37%	X-ray segmentation
Tyagi and Talbar (2022)	LUNA16 (888 scans) Non-public (200 scans)	CT	512 × 512	Architecture	Adv	No	Dice, Sen	80.74%, 76.36%; 85.46%, 82.56%	CT segmentation

tional networks, Zhao et al. (2022) proposed using a Semantic Consistency GAN (SCGAN) for thyroid nodule classification in multimodal medical images. The model overcomes visual differences between different modalities using self-attention mechanisms. Simultaneously, to enhance the semantic consistency of different modal features, nested metric learning is introduced into the semantic information space. Good results were obtained.

Magnetic Resonance Imaging (MRI) can help detect and treat brain tumors earlier and more accurately, improving survival rates. However, classifying brain tumor MR images with the same appearance or structure is very complex and requires professional knowledge to identify tumors. Neelima et al. (2022) Designed an automatic mechanism for tumor classification using MRI. The authors used min-max normalization for data preprocessing and trained using the CAViaR-SPO optimizer based on risk-conditioned auto-regressive values (Fig. 11). Segmentation tasks were performed using GAN methods. The experiments showed that the proposed GAN based on CAViaR-SPO provided 90% segmentation accuracy. Abirami and Venkatesan (2022) Proposed a Generative Adversarial Network based on the Border Collie Firefly Algorithm (BCFA). First, brain tumor images were preprocessed to remove noise. Then, image segmentation and feature extraction were performed using Support Vector Machines (SVMs). Finally, the output results were used to train a BCFA-based GAN for brain tumor severity classification. The experiments achieved good results, with accuracy, specificity, and sensitivity all at 97.515% (see Table 18).

Table 13 Methods that use GAN-based for segmentation (Cardiac)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Wu et al. (2021a)	MICCAI 2009(1) MICCAI 2017(2) (1563+1891 slices)	MRI	n/a	GAN (MDC,MSPN)	Adv odm	No	(APD,ODM,PGC) Endo(1) Epi(1) Endo(2) Epi(2)	1.71, 1.64, 0.928; 0.957, 97.34, 98.21; 1.74, 1.70, 0.930; 0.958, 97.21, 98.13	MRI segmentation
Cui et al. (2021)	MM-WHS (160 images)	MRI CT	n/a	GBCUDA (GAN)	Adv, ce KD Dice	No	Dice	81.5%, 59.2%	MRI,CT segmentation
Le et al. (2021)	MM-WHS 2017 (120 slices)	CT	512×512	R2Unet- GAN	Adv	No	DSC	91.5	CT segmentation
Zhang et al. (2022a)	Non-public (33 subjects)	Cardiac MRI	256×256	TLMDB GAN U-Net	Adv moe	No	Dice	0.9399, 0.9697	MRI segmentation

4.3.3 GANs + other models

The technology based on Generative Adversarial Networks (GANs) can synthesize high-quality medical images and assist in classification tasks, but their classification ability has not been fully explored yet. In Zhou et al. (2021b), a modified GAN framework and a 3D Fully Convolutional Network (FCN) were proposed to improve the classification performance of Alzheimer's disease (AD). T1-MRI scans were used as input data to train the proposed framework. The experiment demonstrated that the modified GAN-based framework effectively enhances the AD classification performance. To help radiologists identify suspicious lesions on mammography X-ray images earlier and reduce mortality, Baccouche et al. (2022) proposed using image-to-image translation models CycleGAN and Pix2Pix to synthesize mammography X-ray images. The generated data was utilized to train the proposed all-in-one fusion model, which is based on the YOLO (You-Only-Look-Once) approach. This model aims to classify breast masses, calcifications, architectural distortions, and normal findings in mammography X-ray images. The experiment results demonstrated that the suggested approach can be used effectively in the field of biomedical applications (see Table 19).

4.4 Detection

Medical image lesion detectors based on DL can effectively help doctors improve diagnostic efficiency and treat diseases in a timely manner. However, accurately identifying abnormalities from images using DL requires a significant amount of labeled training data, which is not readily available. Therefore, it is beneficial to use GANs to generate synthetic images to augment the training set of the detector. Examples of this approach include MSI detection of tissue pathology images of colorectal cancer (CRC) (Krause et al. 2021) and mammography images for hidden cancer detection (MO) (Lee and Nishikawa 2022). In addition, an automated MI detection model called SLC-GAN was proposed in Li et al. (2022b) (Fig. 12). The SLC-GAN model mainly uses GAN to generate high-quality synthetic single-lead ECG data, which is then automatically detected with real ECG data using CNN. A 99.06% accuracy was achieved through 5-fold cross-validation. Krause et al. (2021), Lee and Nishikawa (2022), Li et al. (2022b) all demonstrate through experiments that synthetic images can effectively improve detection performance.

The most basic principle in anomaly detection of medical images is to be able to find abnormal features in lesion images corresponding to normal appearance images. This requires special design to provide useful auxiliary information for medical imaging detection tasks, such as through segmentation (Kumar et al. 2022) or feature extraction (Bhattacharyya et al. 2022; Kumar et al. 2023). In Badashah et al. (2021), a score-Harris Hawks optimization-based GAN (F-HHO-based GAN) model was proposed for early detection of osteosarcoma. In this study, GAN performs osteosarcoma detection by extracting features from images during cell segmentation. Jain et al. (2022) Proposed using a Rider MRFO-based GAN for glaucoma detection (Fig. 13). The Rider MRFO method adopts MRFO technology and Rider optimization algorithm. Since glaucoma detection requires the segmentation of the optic cup from retinal fundus images, Jain et al. (2022) used fuzzy local information C-mean clustering to perform segmentation tasks. Similarly, feature extraction is also one of the most important steps in detecting lung images. Bhattacharyya et al. (2022) First used cGAN to segment the original X-ray images to obtain lung images, and then used

Table 14 Methods that use GAN-based for segmentation (Abdomen)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Shen et al. (2022)	CTLN[43] (4089 images) Non-public (50 images)	CT	256×256	URO-GAN, SAT/VAT	Adv, bce, dice	No	Jac Dic	90.969, 95.208	CT segmentation
Song et al. (2022)	Non-public (391 images) KITS19 (210 patients)	US	512×512	CycleGAN, CNN,U-Net, PSPNet,U-Res, DeepLab v3+	Adv, cyc, dice	No	Acc DSC	0.127, 0.097, 0.105, 0.036; 0.853, 0.850, 0.826, 0.827	US segmentation
Li et al. (2022a)	GlaS (165 images) Prostate Gleason grading (513 images)	Semantic masks, histopathology images	512×512 1024×1024	multi-scale cGAN, VGGNet, m-FCDenseNet	Adv, rec, feat, per	No	GlaS:Dice, F1, Hausdoff mIOU	0.886, 0.894, 61.85 0.721	histopathology images segmentation
He et al. (2022a)	LiTS-2017, KiTS19	CT	$256 \times 256 \times 16$	DCGAN	Adv, ce	No	Dice	0.942 0.942	CT segmentation
Hong et al. (2022)	ISBI 2017/2019 (171 subjects)	MRI, CT	256×256	GAN	Adv, pamr, gt, em	Yes	Dice(DS) Dice(JA) Dice(AC) Dice(PR) Dice(SE) Dice(SP) Dice(ASSD)	0.912 ± 0.037 0.840 ± 0.060 0.991 ± 0.005 0.931 ± 0.024 0.897 ± 0.070 0.997 ± 0.001 0.214 ± 0.239	MRI/CT segmentation
Jafari et al. (2022)	KiTS19(210 subjects) MMWHS 2017(20/20 MRI/CT)	CT, MRI	512×512 256×256	GAN, LMISA, MLNGM	Adv, ce, rec WGAN-GP, encoder-decoder	Yes	kidney:Dice, Pre, ASSD Cardiac:(CT-MRI,MRI-CT) Dice ASSD	0.780, 0.886, 1.082 70.8%, 81.5% 3.6, 2.3	MRI/CT segmentation
Demir et al. (2022)	LiTS (131 scans)	CT	512×512	GAN	Adv	Yes	Dice, recall pre	0.9433, 0.9515 0.9376	CT segmentation

DNN to extract features from the obtained lung images. In the process of medical image detection, it is important to minimize human intervention. Therefore, a skin disease automatic detection algorithm called DAME, different from Bhattacharyya et al. (2022), was proposed in Hu et al. (2022). The algorithm uses a pix2pix-based framework to generate images by learning skin images with melanin and sebum containing structure and context information. Then an enhanced U-Net model is trained using these generated images, which not only performs automatic detection but also marks melanin and sebum. Experimental results demonstrate that compared to other algorithms, DAME achieves better and more robust automatic detection accuracy for melanin and sebum.

Additionally, Zhou et al. (2021c) proposed a Semi-supervised Open Set Domain Adversarial Network (SODA) to address the shortage of COVID-19 viral testing kits by improving the automatic detection of COVID-19 in chest X-rays. SODA aligns data distributions across different domains to tackle the issues of domain shift and small dataset size. Experimental results show that SODA achieves leading detection and classification performance in distinguishing COVID-19 from common pneumonia and provides better localization of pathological changes in chest X-rays.

The use of multi-modal MRI in the detection of hepatocellular carcinoma (HCC) is an important task for HCC evaluation and treatment. However, due to the lack of effective mechanisms and constraint strategies to capture the correlation between multi-modal MRI information, the detection accuracy is relatively low. Xiao et al. (2022) Proposed a TrdAL (task-related adversarial learning framework) to address this issue. In their study, a modality-aware Transformer is used for feature fusion and selection of multi-modal MRI to capture the correlation between them. Furthermore, a new task interaction loss function is introduced to enforce high-order consistency between multi-task labels. The proposed model achieves a accuracy=93.33%, sensitivity=93.15%, specificity= 93.71%, and IoU=82.93% in HCC detection experiments using multi-modal MRI. In addition, there are low accuracy and overfitting problems in lung cancer detection. In Kumar et al. (2023), the empirical wavelet transform method is used to extract features from preprocessed X-ray images, which are then input into a self-attention-based generative adversarial capsule classifier using the sunflower optimization algorithm (SFOA). This approach successfully solves these problems.

4.5 De-noising

In the field of medicine, the accuracy of disease diagnosis often relies on the resolution of radiological imaging. Obtaining high-quality images usually requires a trade-off between image contrast and radiation hazard. High radiation doses can increase contrast to aid in diagnosis, but typically pose radiation risks to patients. Reducing the dose intake can lower radiation exposure but results in lower contrast and lower signal-to-noise ratio. Image denoising is a fundamental field in image processing. Previously, using deep learning to denoise low-contrast images has been an important technical tool to improve their resolution. However, most of the images produced by these techniques are often blurry. With the maturity of GAN technology, it has alleviated many complex problems in image denoising and achieved good results. Using the power of GANs to create lifelike and detailed images, numerous researchers have developed different GAN-based techniques to enhance the quality of images (You et al. 2018). For instance, Zhou et al. (2022) proposes an improved cGAN model for reducing speckle noise in OCT images (Fig. 14). Sun et al. (2022c) proposes the

Table 15 Methods that use GAN-based for segmentation (Spine)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Gong et al. (2022)	Non-public (169 images)	T2-MRI	640×640	ResAtten-GAN	Adv, sce, mce	No	mIOU(%) mPPV(%) SP(%)	89.2 ± 1.8 94.2 ± 2.3 94.4 ± 2.4	T2-MRI segmentation
Zhang et al. (2022b)	Non-public (293 scans)	CT scans	512×512	GANs,2D Vnet,CSA	Adv, ce, Dice WGAN-PG	No	Dice	95.08%, 97.44%	CT segmentation
Li et al. (2022d)	Non-public (69 subjects)	MRI	n/a	LPM-GAN	Adv, bce	No	Recall dice	0.931, 0.904 0.920, 0.903	MRI segmentation

Table 16 Methods that use GAN-based for segmentation (Cell)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Liu et al. (2021a)	Non-public (332 images)	Cell images	333×333	ACAM, cGANs, U-Net	Adv bce Dice	No	Dice	88%	Cell images segmentation
Wang et al. (2022)	MoNuSeg[2] (30 images) PanNuke[3] (45 patches)	Pathological image	1000×1000	RCSAU-Net, GAN	Adv Dice ce	Yes	ACC, AJI F1-Score Dice, HD Recall	0.919, 0.619 0.867 0.820, 4.334 0.895	Nuclei images segmentation
Deng et al. (2022)	BBBC (9600 stained cells) (5 subsets of 20 images per group)	microscope (cell) images	960×960	ELM-GAN (GAN, LFMMG, CMTD)	Adv NHL_2	No	(MCE, RMSE, PSNR, SSIM)	= 0.342, 0.404, 65.99%, 0.961 = 0.803, 1.023, 74.56%, 0.988	Cell images segmentation

[2] https://doi.org/10.1109/TMI.2017.2677499

[3] https://doi.org/10.1007/978-3-030-23937-4_2

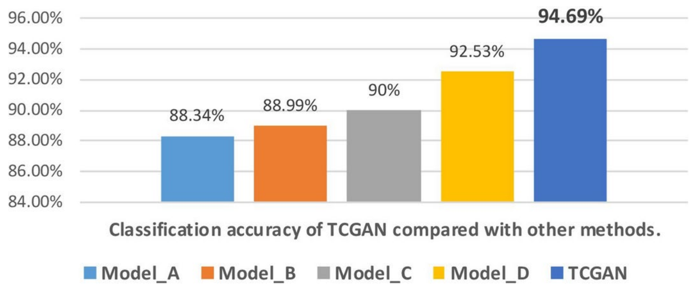

Fig. 10 Xia et al. (2023) proposed a TCGAN method to generate ECG and added it to the original dataset for enhancing ECG classification. In the experiment, the authors used Accuracy as the evaluation metric for classification performance and compared it with four models that did not use GANs. The results showed that the TCGAN method using GAN significantly improved the classification Accuracy to 94.69%. The four models without GAN, namely Model_A, Model_B, Model_C, and Model_D, were obtained from (Zhang et al. 2014; Mar et al. 2011; Yan et al. 2021; Wang et al. 2020), respectively

Table 17 Classification GAN-based methods-(GANs+CNN)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Pang et al. (2021)	Non-public (1447 images)	US	128 × 128	TGAN, CNN	Adv KL	No	Acc Sen Spe	90.41% 87.94% 85.86%	Breast US image Classification
Ma et al. (2022)	MIT-BIH (48 ECG records)	ECG	n/a	ECG-DC-GAN, CNN	Adv ce	No	Acc	98.7%	Arrhythmia ECG image Classification
Barshooi and Amirkhani (2022)	Kaggle, IEEE8023, CheXpert-v1.0-small, Radiology assistant (3604200 images)	X-Ray	968×768	GAN, CNN	Adv bce	No	Acc	98.5%	Chest X-Ray image Classification
Hazra et al. (2022)	Eonelab (12756 images) TCIA (10300 images) Mendeley (10121 cell images)	Cell images from bone marrow aspirate smears	360 × 360	C-WGAN-GP (GAN, CNN)	Adv, WGAN-GP	No	Acc IS	96.98% 14.52	Bone marrow Cell images Classification
Xia et al. (2023)	MIT-BIH (47 patients)	ECG	n/a	TCGAN	Adv	No	Acc	94.69%	Heart ECG image Classification

use of a Pix2Pix model to effectively reduce noise levels by synthesizing low-dose SPECT images from full-dose MP SPECT scans. Similarly, in Gajera et al. (2021), a Generative Adversarial Network (GAN) model is proposed, which uses VGG19 perceptual loss, Wasserstein adversarial loss, and Charbonnier distance structural loss for CT image denoising.

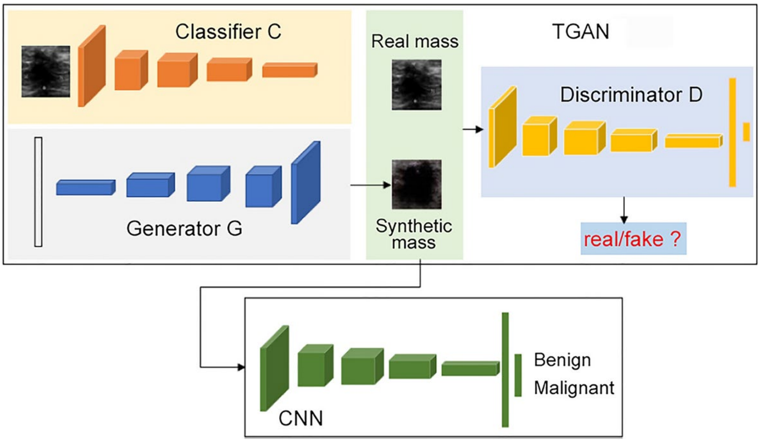

Fig. 11 Proposed framework by Pang et al. (2021) (Classification)

Table 18 Classification GAN-based methods–(GANs)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Lin et al. (2022)	Non-public 1. Heer(6 × 15000 images) 2. Herlev(917 images)	cell images	56 × 56 × 3	SRGAN	Adv, L1 MSE, cyc, ce	No	1 (4, 6 class) 2 (2, 7 class), Acc	0.95%, 93.5% 98.1%, 97.6%	Cervical cell images classification
Zhao et al. (2022)	Non-public (1937 nodules)	US	256 × 256	SCGAN	Adv, vd cond, un-cond, CD-GP	No	Sen AUC Acc Spe	93.43 ± 0.35 97.02 ± 0.57 94.30 ± 0.48 95.14 ± 0.42	Thyroid US image classification
Neelima et al. (2022)	BRATS 2018 (30 patients) Figshare (233 patients)	MRI	n/a	GAN	Adv	No	Acc Sen Spe	91.7% 92.8% 92.5%	Brain MRI classification
Abirami and Venkatesan (2022)	BRATS 2018 (30 patients) Figshare (233 patients)	MRI	n/a	BCFA-based GAN	Adv	No	Acc Sen Spe	97.515% 97.515% 97.515%	Brain MRI classification

Table 19 Classification GAN-based methods–(GAN+other)

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Zhou et al. (2021b)	AIBL, n = 107 NACC, n = 565 ADNI, n = 151	T1 MRI	208 × 240 × 256	GAN, FCN	Adv CEL_1 classifier	Yes	AUC(ADNI) AUC(AIBL) AUC(NACC)	0.932 0.940 0.907	Alzheimer MRI classification

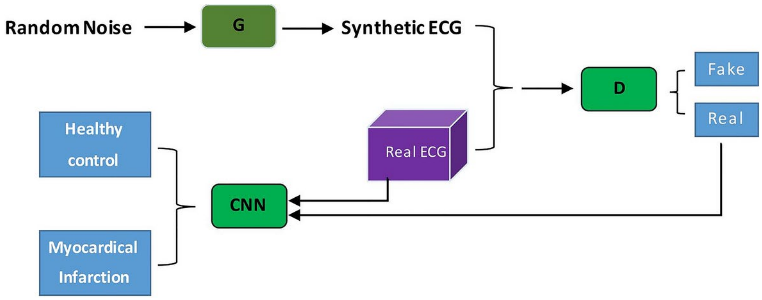

Fig. 12 Proposed framework by Li et al. (2022b) (Detection)

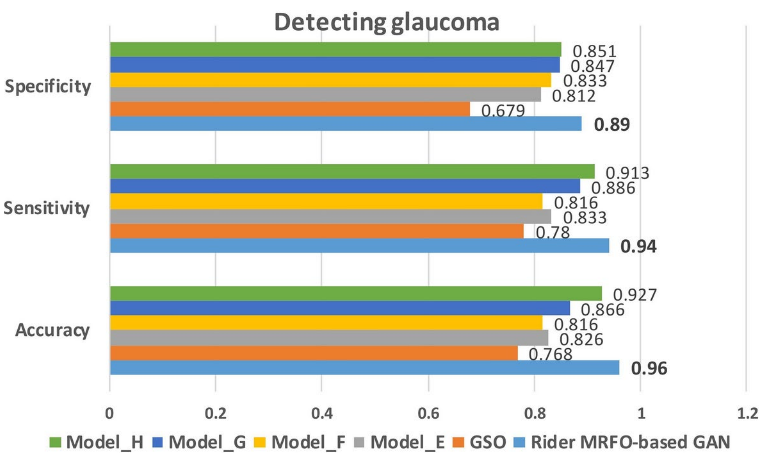

Fig. 13 Jain et al. (2022) proposes a Rider MRFO-based GAN model for glaucoma detection. The authors use Accuracy, Sensitivity, and Specificity as performance evaluation metrics and compare the proposed Rider MRFO-based GAN model with four other models that do not use GANs. Experimental results indicate that the performance of the proposed model surpasses that of the other four models. These four without GAN models are denoted as Model_E, Model_F, Model_G, and Model_H, sourced from (Modified U-net CNN) (Sevastopolsky 2017), (Deep learning with CNN) (Juneja et al. 2020), (based on edge detection) (Septiarini et al. 2017), and (Regions with Convolutional Neural Network) (Bajwa et al. 2019), respectively

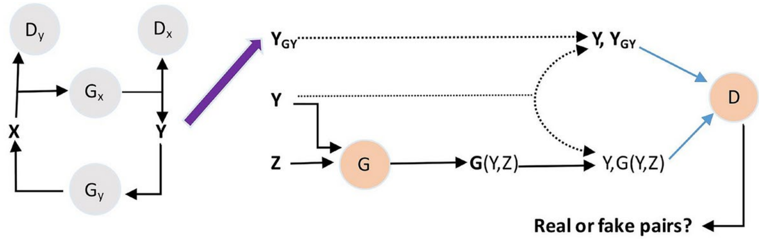

Fig. 14 Proposed framework by Zhou et al. (2022) (Denoise)

Huang et al. (2022a) employs a DU-GAN denoising method with a U-Net framework as the discriminator to better address the issues of noise and artifacts that may arise in CT scan images obtained at low radiation doses. By learning the most discriminative local and global differences between normal-dose images and denoised images, it effectively improves the quality of LDCT images.

Currently, GAN-based image denoising techniques are receiving increasing attention and achieving good results. Unfortunately, many current denoising algorithms are very complex, and the memory requirements and parameter count pose many inconveniences for model training. Therefore, Wang and Hu (2021) proposed a new progressive WGAN model that aims to use a simpler network to denoise LDCT images. The model introduces a recursive algorithm to reduce network parameters. At the same time, the model incorporates a weighted structure-sensitive hybrid loss (WSHL) function to reduce artifacts and preserve more details in the LDCT image denoising process. Wang et al. (2021a) uses a small parameter capsule cGAN (Caps-cGAN) approach to enhance the ability to capture complex and sparse features in OCT images, while reducing a large number of parameters. The model also incorporates a new and unique loss function that combines both supervised and unsupervised learning. This helps prevent the loss of important structural information in the retina during training. Successful denoising of retinal OCT images was achieved. In addition, Gu and Ye (2021) not only reduced additional memory requirements and parameter count, but also obtained better LDCT image denoising performance compared to previous CycleGAN methods. In the process of organ screening and disease diagnosis, some medical images are often damaged by metal artifacts. This affects the doctor's judgment of the anatomical structure and may threaten the patient's life. In order to reduce the impact of artifacts and improve image quality, Xu et al. (2022b) proposed a GAN-based MAR-GAN model for denoising maxillofacial CT images, which not only preserves anatomical structures but also restores detailed structural information in low-quality images. Meanwhile, Huang et al. (2022b) proposed a new GAN-based method for reducing metal artifacts (Fig. 15). Using mutual information and CT images to combine into a comprehensive feature representation, generating multimodal feature fusion. In addition, the authors also introduced an edge enhancement subnetwork in the model to effectively avoid secondary artifacts and suppress noise. Bone suppression is also considered an image denoising problem because lung bones can affect the localization of lesions or nodules in chest X-rays (CXR). Currently, many cGAN-based models are used to generate bone suppression images from CXR, but they do not focus on preserving spatial features and image information. Therefore, Rani et al. (2022) proposed a spatial feature and resolution maximization (SFRM) GAN, which incorporates both Sobel and Perceptual loss. The model not only maximizes the preservation of

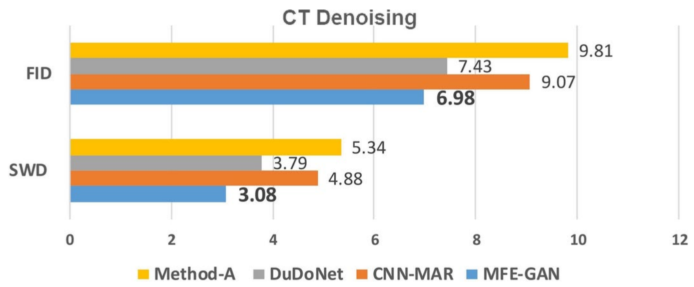

Fig. 15 Huang et al. (2022b) propose using Edge-Enhanced Generative Adversarial Networks for multi-modal feature fusion to reduce CT metal artifacts. Experimental results compare three denoising methods without using GANs. From the evaluation metrics FID and SWD, it's evident that the proposed MFE-GAN model outperforms others. Method-A from Kalender et al. (1987)

image quality and intrinsic information, but also effectively improves bone suppression by reducing the visibility of bones in CXR. The model achieved a confidence level of 0.95 in experiments. Rani et al. (2022) Demonstrated that the model can be used for image denoising and pre-processing.

4.6 Reconstruction

In the medical field, as computer technology continues to advance, medical image reconstruction techniques play an increasingly important role in clinical applications. Researchers are particularly interested in the capabilities of GANs, which can generate realistic images. GANs are able to reconstruct high-resolution images from low-resolution ones and effectively reduce image artifacts You et al. (2020b). Most current image reconstruction techniques involve 2D, 3D, and 4D imaging.

4.6.1 2D images

High-resolution (HR) MRI images help diagnose brain diseases. However, obtaining HR MRI is largely limited by cost and time. Therefore, Huang et al. (2022c) Proposed using Parallel Imaging Coupled Double Discriminator GANs (PIDD-GAN) to reconstruct high-quality brain MRI. This study introduced content loss in the generator to improve image reconstruction quality. Meanwhile, using double discriminators aims to enhance the MRI edge information reconstruction, which can be used for both overall brain MRI reconstruction. Through comprehensive experimental results on public datasets, PIDD-GAN is shown to be able to reconstruct high-quality MRI with intact edge information. In the process of MRI reconstruction, not only image quality but also network complexity and reconstruction time need to be considered. In order to effectively improve the accuracy of brain disease diagnosis and reduce radiation hazards, in addition to reconstructing high-quality MRI Huang et al. (2022c), Xue et al. (2022) proposed a deep learning-based cGAN framework to reconstruct standard dose PET imaging. In addition, Fu et al. (2022) proposed the transGAN-SDAM method for reconstructing high-quality full-dose brain PET images from low-dose PET images.

Craniofacial deformities can cause serious psychological problems for patients. In the medical field, preoperative virtual design methods are usually used to help accurately repair craniofacial tissue and organs, which is the main method. However, traditional preoperative virtual image reconstruction technology is relatively time-consuming, complex, and not universally applicable. To solve the above problems, Xu et al. (2022c) proposed using RecGAN based on deep learning methods to intelligently assist preoperative reconstruction of craniofacial deformities for patients (Fig. 16). The model was trained using a self-built craniofacial CT dataset. The proposed model can assist in reconstructing normal bone tissue morphology of craniofacial deformity patients in a more universal and personalized manner.

4.6.2 3D/4D images

In medical image reconstruction tasks, most algorithms for reconstructing high-resolution images are usually used to perform 2D image operations. For example, Kim et al. (2022) proposed using a GAN network to detect anomaly scores (AS) to reconstruct breast images of breast cancer patients. The model performs standardization on the input normal breast images using z-scores. Then, a tissue expander is used to reconstruct the breasts of breast cancer patients who have undergone breast removal surgery. The experiments have demonstrated that the suggested approach produces excellent aesthetic outcomes for breast reconstruction. In addition, reconstruction tasks for high-resolution 3D and 4D images are also used in medical analysis systems. For example, Jiang et al. (2021) proposed using a Fusion Attention Generative Adversarial Network (FA-GAN) to reconstruct high-resolution 3D MRI from low-resolution MRI. The authors used global feature fusion and local feature fusion modules in the model to enhance the important features of the MRI and extract them. Ghodrati et al. (2021) Proposed using 3D GAN technology to reconstruct 4D cardiac MR images. The GAN model maintains anatomical accuracy and temporal consistency during the training process using a progressive growing strategy to improve the time and quality of image reconstruction.

4.7 Fusion

Currently, medical imaging has become the main auxiliary tool for disease diagnosis. Due to the lack of information in certain fields, diagnosing diseases using a single mode has become a dilemma. With the continuous development of computer technology, image fusion technology based on GAN has been used to address this problem. Medical image fusion refers to the merging of multiple images captured through different imaging modes into a single integrated image. The goal is to generate a specific organ fusion image with significant features and improved quality. The fusion image contains information from multiple modal images, which facilitates diagnosis and segmentation (Liu et al. 2022a; Abirami et al. 2022).

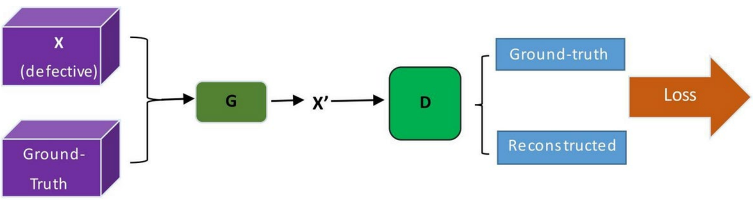

Fig. 16 Proposed framework by Xu et al. (2022c) (Reconstruction)

Currently, numerous researchers have put forward different image fusion methods and achieved positive outcomes. Nevertheless, many fusion techniques still face challenges in terms of slow speed and unclear details. To address this issue, Fu et al. (2021) proposed a GAN framework based on dual-stream attention mechanism (Fig. 17). The framework uses dual-stream architecture and multi-scale convolution to extract detailed features. Furthermore, an attention mechanism is employed to amplify the fusion of these features. The training process continues until both the generator and discriminator reach a Nash equilibrium. The outcome demonstrates that the proposed model achieves superior fusion results with reduced time consumption. Moreover, many current image fusion methods struggle to preserve crucial information and overlook the most distinctive regions of the image. Therefore, to address these issues, Wang et al. (2022e) proposed an Attention-guided Multi-model Fusion Network (AMFNet) for multi-model image fusion. The model uses attention networks for remote dependency and information refinement networks to obtain internal representations and feature maps of the image, respectively. Additionally, the model incorporates a network that combines the attention network and information refinement network. Moreover, a special module called the convolutional block attention module is added to the discriminator. This module helps the discriminator focus on the most important parts of the multi-modal source image. The research results prove that the AMFNet method can retain more image detail information and outperform other methods in visual effects. In order to accurately retain the functional and structural details of the original image, the authors of Fan et al. (2023) introduced a comprehensive model called U-Patch GAN. This model, based on GAN, enables the self-supervised merging of different types of brain images. To enhance the attention of the generator (classic U-net network) to high-frequency information, the model introduces a dual-adversarial mechanism based on Markov discriminator (PatchGAN). An adversarial and feature loss function based on F-norm is also proposed, which significantly improves the quality of the fusion image.

4.8 Registration

In the process of medical image analysis, multiple images of the same patient are strictly aligned and analyzed together to obtain comprehensive information for the purpose of effectively improving medical diagnosis and treatment. Experiments in previous studies have shown that CNN can be used to align medical images efficiently using a single pass through the network. With the continuous development of deep networks, GAN has been increasingly used by researchers to extract better matching mappings. For example, Sood et al. (2021) proposed the use of a novel multi-image super-resolution GAN (miSRGAN) (Fig. 18) to register reconstructed 3D prostate tissue pathology volumes to reconstructed 3D MRI (Fig. 19). However, many current methods for aligning images, particularly those

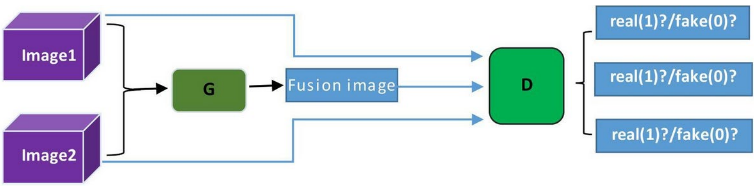

Fig. 17 Proposed framework by Fu et al. (2021) (Fusion)

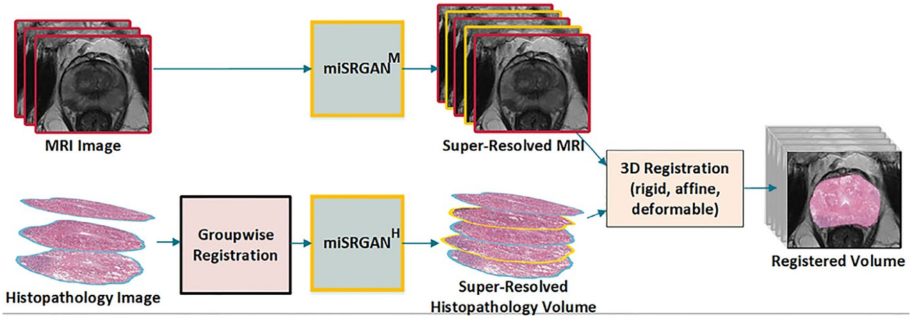

Fig. 18 Proposed framework by Sood et al. (2021) (Registration)

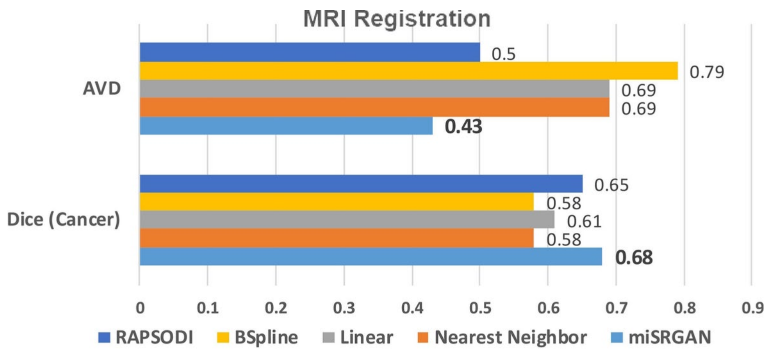

Fig. 19 Sood et al. (2021) introduced miSRGAN, a model designed for 3D registration of prostate tissue pathology images and MRI. Experimental results were compared with four registration methods that do not use GANs. Based on evaluation metrics AVD (Absolute Volume Difference, smaller is better) and DICE (Dice Similarity Coefficient, larger is better), we can conclude that the performance of registration methods without GANs is inferior to miSRGAN

involving different types of images, face limitations such as slow optimization processes, challenges in understanding connections between different types of images, or the high expense of manually labeled data. To address these issues, Zheng et al. (2022) proposed a novel symmetric image registration method SymReg-GAN based on GAN and trained it with a semi-supervised strategy. It effectively eliminates time-consuming iterations and directly generates registered images, making valuable labeled and unlabeled data fully utilized. Aligning different types of medical images is a crucial process in various medical image analysis tasks. Recent developments in aligning multimodal medical images depend on transforming one type of image into another to achieve better results. However, the performance is still restricted because of the inadequate utilization of complementary techniques between image alignment and transformation, which can enhance the accuracy of both processes simultaneously. In order to achieve this, Lian et al. (2022b) introduced a new approach called CoCycleReg that combines image registration and translation in a collaborative and consistent manner. They introduced a unique registration network with two heads to extract features and predict deformation fields. The experiments conducted on T1-T2 (MRI) and CT-MRI datasets demonstrated that CoCycleReg outperformed other conventional and deep learning methods that are currently considered the best in the field. In the study by Zhu et al. (2022), they introduced a two-stage generative adversarial network (GAN) method for registering multimodal brain images. They used U-Net as the core of the GAN. In the first stage, the GAN takes the image structure created by the improved MIND as input and generates the predicted structure of the registered image. In the second stage, the GAN estimates the intensity of the registered image using the structural representation obtained in the first stage. The proposed method demonstrated promising outcomes.

In clinical diagnosis, the use of 18F-sodium fluoride (18F-NaF) positron emission tomography (PET) in imaging coronary artery disease activity has shown great potential. However, the current method of manually aligning PET and computed tomography (CT) angiography data is subjective and leads to subjective image processing. To address this, Singh et al. (2022) proposed a fully automatic angiography registration framework using cGAN and nonlinear diffusion homomorphic (devil) registration. The generated pseudo-CT is used to automatically register PET to CT angiography. This eliminates the subjective influence of humans on image registration (see Tables 20, 21, 22, 23, 24).

4.9 Prediction

In the current era, clinical prediction models play a crucial role in the field of medicine. Prediction models primarily use fully or (semi/non-)parametric mathematical models to calculate the probability risk of an individual already having a certain disease or the possibility of causing a certain outcome in the future. Clinical prediction models can be roughly divided into two types: diagnostic models and prognostic models. For example, Wu et al. (2021b) demonstrated through experiments that deep learning-based prediction model algorithms using CNN and GANs can achieve higher performance in reconstruction and diagnostic accuracy. These models not only effectively assist disease diagnosis and reduce repeated examinations but also alleviate the economic burden of patient checkups. In addition, Uemura et al. (2021) developed a pix2surv model to quickly and accurately estimate disease progression and mortality. This model directly estimates event occurrence time information from the patient's chest CT images for survival analysis.

4.9.1 Diagnosis

The purpose of diagnostic models is to predict the probability risk of a patient having a certain disease based on the manifestation of certain symptoms. However, manual analysis of medical images is highly subjective, leading to poor consistency and low sensitivity in clinical diagnosis. Hence, Qi et al. (2022) devised a computerized system for diagnosing breast cancer, which employs ultrasound images to enhance the precision of breast cancer detection. When it comes to brain conditions like Alzheimer's disease (AD), PET scans can detect cellular transformations in organs and tissues earlier than MRI scans. Nevertheless, the availability of PET data is often limited due to cost, radiation concerns, and other constraints. To address this problem, Zhang et al. (2022d) proposed a 3D end-to-end GAN network framework using U-Net as the generator architecture (Fig. 20). The model transforms MRI scans into PET scans that are more similar to ground truth using 3D GP loss and SSIM loss, aiming for multimodal Alzheimer's disease diagnosis. Currently, computer tomography angiography (CTA) is a more effective method than ordinary CT in the diagnosis of

Table 20 Methods that use GANs for detection

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Badashah et al. (2021)	Reference (1144 images)	H&E	1024 × 1024	F-HHO-based GAN	Adv	No	Acc, Sen, Spe	98%, 98% 98%	Osteosarcoma H&E assisted detection
Li et al. (2022b)	PTBDB (200 MI subjects, 8377 samples)	ECG	n/a	SLC-GAN, CNN	Adv BCE	No	Acc	99.06%	MI ECG image assisted detection
Jain et al. (2022)	(45 images)[4]	Fundus images	884 × 580	Rider MRFO-based GAN	Adv ce, bc	No	Acc, Sen, Spe	0.96, 0.94 0.89	Fundus image assisted detection
Bhattacharyya et al. (2022)	(247 images)[5] (342 images)[6] (1030 images)[7]	X-ray images	2048 × 2048	C-GAN DNN ML	Adv cce	No	Acc	96.6%	Lung X-ray image assisted detection
Hu et al. (2022)	Non-public (720 melanin & sebum images)	Skin images	n/a	DAME (pix-2pix, U-Net)	Adv L_1	Yes	PSNR SSIM F-measure	(33.13, 0.63, 0.71); (34.23, 0.59, 0.63)	Skin image assisted detection
Xiao et al. (2022)	Non-public (135 subjects)	MR images	n/a	TrdAL (GAN, CNN)	Adv, reg RPN, ce	No	Acc, IoU	93.33% 82.93%	Lung MRI assisted detection

4 https://www5.cs.fau.de/research/data/fundus-images/

5 http://www.isi.uu.nl/Research/Databases/SCR/

6 https://github.com/ieee8023/covid-chestxray-dataset

7 https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia/metadata

aortic dissection during clinical diagnosis. However, processing CTA images has disadvantages such as the need for contrast agents, complex and time-consuming procedures, etc., which pose challenges for CTA examination. Therefore, in the study conducted by Chen et al. (2022b), they proposed a two-step approach for generating aortic dissection CTA images. First, they aligned CT images and used an nnU-Net segmentation network to extract only the aorta, resulting in paired CT and CTA images. Then, they employed a cascade GAN method to synthesize the aortic dissection CTA images. To further enhance the quality of the generated images, the authors utilized a cascade generator and dual discriminator network with the DCT channel attention mechanism. The experimental results demonstrated that the developed model significantly improved the accuracy and efficiency of aortic dissection diagnosis, leading to a reduced missed diagnosis rate.

In the field of computer vision, medical imaging technology plays a crucial role in improving disease monitoring. However, deep learning models encounter challenges in

Table 21 Methods that use GANs for De-noising

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Gajera et al. (2021)	Kaggle (75 patients), TCIA	CT images	n/a	CL-GAN	Adv, Per, Charbonnier	No	SD noise	41.96	Alzheimer CT image denoising
Huang et al. (2022a)	2016 LDCT Grand Challenge[8] Non-public (850 CT scans)	LDCT images	512×512	GAN U-Net RED-CNN	Adv Pixel-wise	Yes	PSNR SSIM RMSE	22.3075 0.0802 0.7489	CT image denoising
Wang and Hu (2021)	Non-public (train 4554 images)	CT images, LDCT images	512×512	WGAN CNN FCN	Adv, L_1, Structural	No	PSNR SSIM RMSE	24.8728 0.8052 0.0571	CT image denoising
Wang et al. (2021a)	Non-public (1920 B-cans)	OCT Images	512×992	cGAN CNNs	Adv,L_1, BCE sem SSIM	Yes	SNR CNR ENL EPI	59.01 11.37 417.22 1.03	OCT image denoising
Gu and Ye (2021)	(40065 images)[9] TCIA (16648 images)	CT images, LDCT images	128×128	Cycle-GAN, U-Net, AdaIN	Adv, disc, cyc, id	No	PSNR SSIM	30.8730 0.6605	Cardiac CT image denoising
Huang et al. (2022b)	DeepLesion (8000 images) Non-public (50 CT images)	CT images	256×256	MFE-GAN, CNNs	Adv feature	No	IFC PSNR SSIM MSIM FSIM	2.35 31.25 0.939 0.990 0.953	CT image denoising
Rani et al. (2022)	JSRT (247 radiographs)	Chest radiographs	$512 \times 512 \times 3$	pix2pix, WGAN	Adv, L_1, Sobel, Per	No	NMSE SSIM PSNR	0.00025 0.989 43.588dB	Chest radiographs denoising

8 https://aapm.onlinelibrary.wiley.com/doi/abs/10.1002/mp.14594

9 https://aapm.onlinelibrary.wiley.com/doi/abs/10.1002/mp.13284

terms of limited availability of diverse datasets for specific diseases and the difficulty in extracting meaningful and distinctive features. To address this issue, Mostafiz et al. (2022) proposed using a GAN-based approach to synthetic images of COVID-19 and other chest infections, solving the problem of limited data storage for COVID-19 and the inability to provide important features. Then, a faster R-CNN model was used to diagnose COVID-19 from CXR datasets in a multi-class environment with other lung infections. The experimental results achieved a satisfactory diagnosis accuracy of 99.16%. Furthermore, Qu et al. (2022) introduced an innovative approach called generative replay strategy to tackle the issue of data heterogeneity in collaborative learning methods. They used GAN to learn the aggregation of knowledge from all local institutions to regulate the model's training towards an unbiased target distribution. The results from the experiments showed that the proposed

Table 22 Methods that use GANs for Reconstruction

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Jiang et al. (2021)	MICCAI2013 grand challenge (50 samples)	3D MRI	n/a	GAN GFFB LFFB	per pixel Adv	No	PSNR SSIM	42.07,30.28, 34.28 0.9974,0.9926, 0.9966	Brain,Knee cardiac MRI Reconstruction
Ghodrati et al. (2021)	Non-public (42 patients)	MRI (Cine cardiac)	$64 \times 64 \times 64$	TAV-GAN	Adv TA pixel-wise content	No	SSIM nRMSE	0.785 0.030	Cardiac MRI Reconstruction
Huang et al. (2022c)	Calgary Campinas (15360 images) MICCAI 2013 (18850 images)	T1w MRI	n/a	PIDD-GAN	Adv MSE mask Per	No	PSNR SSIM,FID MOS-(Holistic, Edge)	32.4809 0.9438,15.38 3.80 3.40	Brain MRI Reconstruction
Xu et al. (2022c)	Non-public (SMD&CMD) (600 cases), Public(SD 189 cases)	CT images	512×512	RecGAN, cGAN	Adv MSE Per Tar ce	No	PSNR SSIM	31.0534 0.9780	Brain CT image Reconstruction
Kim et al. (2022)	Non-public (61 patients)	CT images	500×500	f-AnoGAN	Adv MSE fs	Yes	(Pre-RT, Post-1Y, Post-2Y) AS	1.99 2.92 2.94	Breast CT image Reconstruction

method enhances the accuracy of predicting diabetic retinopathy and decreases the mean absolute error in bone age estimation.

4.9.2 Prognosis

Prognostic models are different from diagnostic models. After determining the initial state of the patient (diagnosed with a certain disease), prognostic models track the patient's survival status and calculate whether the patient will experience a certain event in the future (such as death). For example, in order to improve the accurate prognosis of glioblastoma multiforme (GBM), a prognostic method for overall survival (OS) based on radiogenomic was developed Islam et al. (2021). First, cGAN was proposed to generate synthetic missing MRI modalities. Then, a modified FCN architecture was used to segment the synthesized MRI and compared with models in segmentation and OS prediction. Finally, it was experimentally demonstrated that adding missing MRI modalities can improve segmentation prediction. Currently, numerous clinical predictive models have achieved good results in predicting the prognosis of patient's diseases in the medical field Xu et al. (2022a). However, pattern collapse still exists in the model training process. Therefore, Zhang et al. (2022e) proposed a prognostic model based on CGAN, PregGAN (Fig. 21). The model uses clinical data and noise vectors of patients to generate prognostic prediction results (Fig. 22). At the same time, gradient penalty strategy and Wasserstein distance are used to address the issue of pattern collapse in the training process (see Tables 25, 26).

4.10 Image enhancement

Medical image enhancement techniques aim to improve the quality of medical images, making them easier for medical professionals to analyze and interpret, thereby enhancing the accuracy of medical diagnoses and analysis efficiency. Despite modern medical imaging devices being capable of generating high-resolution images, in practical applications, these images often suffer from noise, blur, and other factors, leading to a decline in image quality and making the diagnostic process challenging. Therefore, medical image enhancement techniques have become essential tools to address this issue. These techniques include traditional image processing methods such as filtering, enhancement, and denoising, as well as advanced techniques based on deep learning, such as Generative Adversarial Networks (GANs). By applying these techniques, the visual quality, structural details, and contrast of medical images can be improved, providing medical professionals with clearer and more accurate images, thereby contributing to improved diagnostic and analytical outcomes.

Compared to traditional image processing methods, deep learning techniques based on GANs have shown more significant effects in medical image enhancement (Bai et al. 2022). For instance, Yuan et al. (2022) proposed a deep learning-based super-resolution vascular angiography image reconstruction GAN (SAR-GAN) to enhance the quality of wide-field retinal optical coherence tomography angiography (OCTA) images. SAR-GAN addresses the issue of low resolution in OCTA images caused by insufficient lateral sampling and has achieved remarkable results. The study results demonstrate that SAR-GAN generates clearer and more coherent images, significantly improving image quality, especially in terms of noise intensity, contrast-to-noise ratio, and vascular connectivity (Fig. 23). Compared to traditional and other deep learning methods, SAR-GAN exhibits superior image

Table 23 Methods that use GANs for Fusion

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Fu et al. (2021)	The Whole Brain Atlas (120 pairs)	MRI PET SPECT images	256 × 256	DSA-GAN	Adv Content Per SSIM	Yes	Q-AG Q-EN Q-NIQE	7.6204,5.1134 4.7652; 7.0994,5.4984 4.4753	Brain MRI PET SPECT image fusion
Wang et al. (2022e)	Harvard Medical (9984 image pairs)	PET MRI	84 × 84	GAN, IRN, SA	Adv content	No	PSNR SF SSIM VIFF	62.561 0.0590 0.828 0.703	Brain MRI PET image fusion

enhancement performance across various field-of-views, indicating tremendous potential for clinical assessment. Additionally, Sun et al. (2022a) proposed a deep learning method called Cell Image Enhance GAN (CIEGAN), aimed at enhancing microscopic cell images to address the issue of image blur during long-term live cell imaging. This method effectively improves image clarity and facilitates subsequent analysis. CIEGAN can compensate for image blur during critical moments of imaging, while achieving a balance between imaging speed and image quality. Furthermore, the method performs well in enhancing fluorescence images and has been validated for its effectiveness in human-induced pluripotent stem cell-derived cardiomyocyte differentiation experiments, significantly enhancing spatial resolution of images.

In the field of medical image enhancement, researchers have proposed various innovative methods to address issues related to image quality, preservation of structural information, and details (Pan et al. 2023). Firstly, addressing the challenges of traditional methods that require paired high/low-quality image training and the inability to handle unpaired images, some studies have introduced new approaches based on Generative Adversarial Networks (GANs). For instance, FS-GAN proposed by Yu et al. (2023) utilizes a fuzzy self-guided structure-preserving module and illumination distribution correction module to handle unpaired images. Additionally, to improve the quality of medical images and preserve structural information, a fuzzy discriminator is introduced to enhance model performance. Similarly, to tackle the issue of non-paired medical image enhancement in medical research, Xu et al. (2023) proposed a method called SSP-Net based on a Siamese structure with dual-input mechanism. This method incorporates the mechanism of generative adversarial networks. Through collaborative iterative adversarial learning, SSP-Net achieves enhanced medical image structure preservation. Comprehensive experimental results demonstrate the superiority of SSP-Net over other state-of-the-art techniques in non-paired image enhancement.Furthermore, to achieve higher accuracy in computer-aided cancer diagnosis, Khan et al. (2023) introduced Multi-level GAN to enhance CT images. This method demonstrates outstanding performance across multiple datasets, validating the advantages of multi-level GANs in medical diagnosis. Additionally, addressing the issue of low-quality medical endoscopy images, Fan et al. (2022) proposed a method based on conditional entropy generative adversarial networks. Through this method, the quality of medical endoscopy images is significantly improved while preserving image details. Moreover, to enhance the

Table 24 Methods that use GAN-based for Registration

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Sood et al. (2021)	TCIA(315 cases) Non-public (218 cases)	3D Histo MRI	224×224	miSRGAN, VGG	Adv MSE per	No	Dice(Cancer) AVD(mm)	0.68 ±0.06 0.43 ±0.11	Prostate MRI registration
Lian et al. (2022b)	BraTS dataset (285 cases) neck&head dataset (151 pairs)	(MRI) T1-T2 CT images	$80 \times 144 \times 112$	CycleGAN PatchGAN	Adv sim cyc smoothness	Yes	(T1-T2, T2-T2, MRI-CT, MRI-CT) DSC,HD95	90.0%,4.84 89.72%,4.74 74.27%,6.36 74.36,6.45F	Brain MR CT image registration
Zhu et al. (2022)	Atlas(MR,MR-SPECT) (9600 images) BrainWeb,RIRE (12000 images)	3D MR SPECT	$128 \times 128 \times 64$ $128 \times 128 \times 128$	TGAN, U-Net	Adv MSE LS MSSIM	No	(Atlas, BrainWeb) Dice	0.76 0.84	Brain MR SPECT image registration
Singh et al. (2022)	Non-public (169 patients)	PET CT pseudo-CT	256×256	GAN, Unet, CNN	Adv mean-squared	No	TBR,SUV	0.31,0.26	Coronary PET CT image registration

quality and diagnostic accuracy of retinal images, Wu et al. (2023) introduced Semi-Supervised GAN with anatomy structure preservation (SSGAN-ASP). This method enhances images by introducing anatomy structure extraction components and retaining color information, thereby improving visual quality and clinical diagnostic performance. Additionally, addressing common issues in medical image enhancement such as uneven illumination distribution and loss of texture details, Zhong et al. (2023) proposed Multi-Scale Attention Generative Adversarial Networks (MAGAN). This method enhances image quality more significantly through feature pyramids and attention distribution mechanisms. These related works have made significant advancements in various fields, providing important references and insights for the development of medical image enhancement technology.

Despite the existence of many GAN-based image enhancement methods that provide satisfactory visual results, they often face the trade-off between maintaining originality and image quality. Therefore, Cap et al. (2023) proposed an innovative unsupervised medical image enhancement framework named Laplacian Medical Image Enhancement (LaMEGAN) to address this issue. The LaMEGAN framework can extract more original and high-quality information from low-quality medical images. This method demonstrates superior performance in capturing changes in image structure. Experimental results confirm that LaMEGAN can generate images with extremely high quality while preserving image structure, successfully achieving a balance between quality and originality (see Table 27).

5 Overview

Research on GAN in medical image computing has been growing rapidly, reflecting the increasing interest in the application of GAN in the medical imaging domain.

5.1 Key aspects of successful learning method

In the field of medicine, radiation oncologists typically use medical images as important tools for diagnosing diseases and administering radiation therapy. However, obtaining medical images has always been a challenging task due to issues such as patient privacy protection, scanning costs, radiation exposure, and scan time, resulting in a severe lack of and imbalance in medical image data. Even though there are now many public datasets available, datasets for training deep learning models are still relatively insufficient, especially datasets that require annotations by professional doctors which are time-consuming and difficult to collect. In the past few years, numerous researchers have utilized GAN frameworks as medical image synthesis methods (Sect. 4.1) to generate realistic medical images, effectively helping to address the lack of clinical datasets and the problem of dataset imbalance, expanding the clinical medical image dataset required for training. Good success has also been achieved in cross-modal medical image-to-image translation tasks, such as MRI $\rightarrow$ CT, PET $\rightarrow$ CT, etc. In terms of cross-modal synthesis, Cycle GAN model can generate high-quality synthetic images without the need for paired training data, and uses cycle consistency constraints to control the consistency of the generated images. It perfectly solves the restriction that training data must be paired.

In addition, high-quality medical images can increase the accuracy of disease diagnosis during the diagnosis process. However, obtaining high-quality images often requires

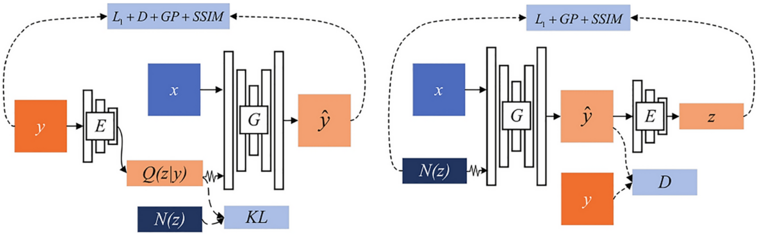

Fig. 20 Proposed framework by Zhang et al. (2022d) (diagnosis)

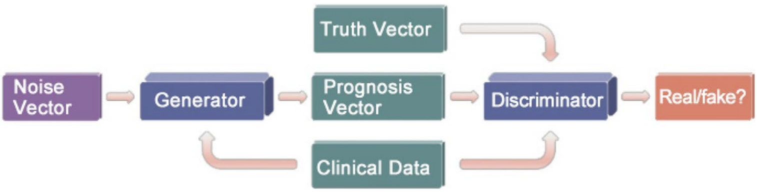

Fig. 21 Proposed framework by Zhang et al. (2022e) (Prediction)

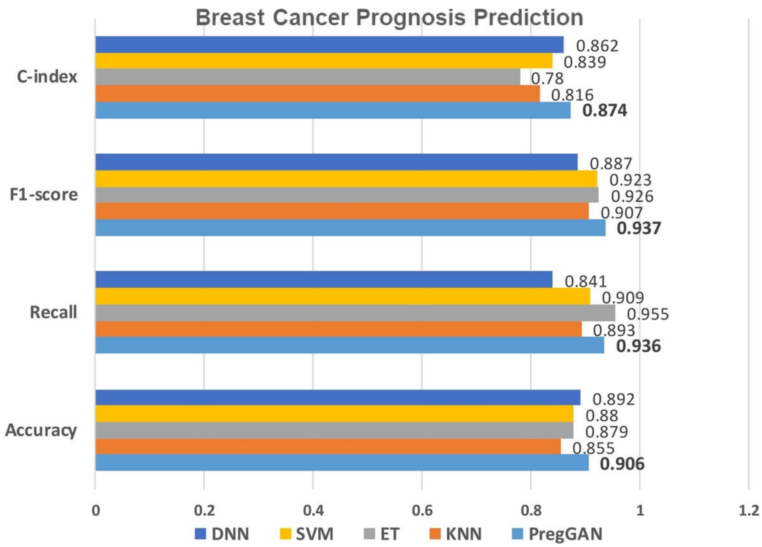

Fig. 22 Zhang et al. (2022) a prognosis prediction model for breast cancer based on the PregGAN model is proposed. The authors compare its performance with four traditional prognosis prediction models using metrics such as C-index, F1-score, Recall, and Accuracy. Experimental results demonstrate that the use of GANs in the PregGAN model outperforms traditional methods without GANs in prognosis prediction

a trade-off between image contrast and radiation hazard. High radiation doses can often pose radiation hazards to patients, and reducing the dose intake can result in lower contrast and lower signal-to-noise ratio in the images. Using deep learning to denoise low-contrast images typically produces images that are mostly blurred. With the maturity of GAN tech-

Table 25 Methods that use GAN-based for prediction-diagnosis

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Zhang et al. (2022d)	ADNI (1732 image pairs from 873 subjects)	PET MRI	$128 \times 128 \times 128$	3D GAN	GP,L_1 SSIM adv KL	No	Acc	85.00% 56.47%	Brain PET MRI assisted diagnosis
Chen et al. (2022b)	Non-public (1114 images)	CT CTA images	n/a	cascaded GAN	Adv	No	PSNR SSIM	32.98 0.9905	Aortic CT/CTA image assisted diagnosis
Mostafiz et al. (2022)	(3933 images)	X-ray	224×224	GAN,Faster R-CNN	Adv	No	Acc	99.16%	Chest X-ray image assisted diagnosis
Qu et al. (2022)	Retina (17563 pairs images) BoneAge (5575 images)	fundus images	256×256	GAN, VQ-VAE-2	Adv MAE	No	Acc(%) PSRN	77.20 ± 0.13 11.82 ± 1.86	Fundus image assisted diagnosis

10 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4256233/

11 https://link.springer.com/article/10.1007/s12065-020-00550-1

Table 26 Methods that use GAN-based for Prediction-Prognosis

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Islam et al. (2021)	BraTS-2017 (331 cases) TCGA (202 cases)	MRI Genomic	240×240	cGAN FCN	Adv	No	Acc	84.62%	Brain MRI Genomic assisted prognosis
Zhang et al. (2022e)	METABRIC (490,1490) TCGA-BRCA (348)	clinical traits, TCGA	256×256	Preg-GAN	Adv GP Wasserstein	Yes	ACC AUC	90.6% 0.946	Breast clinical traits TCGA assisted prognosis

nology, it has alleviated many complex problems in image denoising and achieved good results. Taking advantage of the ability of GANs to generate realistic and high-resolution images, many researchers have proposed various GAN-based methods to denoise images with higher perceptual quality (Sect. 4.5).

Compared to ordinary images, medical images have unique patterns that are crucial for disease diagnosis. During the process of training GAN models using medical images, the discriminator learns significant features extracted from the training data to iteratively update the generator, achieving model optimization. GANs have a significant advantage in extract-

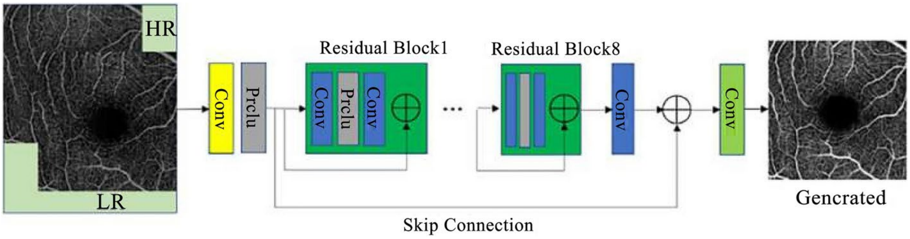

Fig. 23 The image enhancement framework proposed by Yuan et al. (2022)

Table 27 Methods that use GAN-based for Image Enhancement

References	Dataset			Model overview			Evaluation		Application
	Name & quantity	Modality	Resolution	Architecture	Loss	Code	Metric	Performance	Remark
Yu et al. (2023)	CORN-2 (688 images)	Confocal microscopic images	384 × 384	FS-GAN (U-Net, SSRM, IDCM)	Adv, tf cyc, id	No	AUC Acc Sen Kappa Dice G-mean	0.731±0.048 0.988±0.001 0.518±0.097 0.451±0.183 0.452±0.095 0.713±0.084	Confocal microscopic image enhancement
Khan et al. (2023)	Ircadb Sliver07 LiTS (190 subjects)	CT	128 × 128	Multi-level GAN	Adv, content	No	PSNR SSIM	13.0007 18.013,17.61 0.335 0.54,0.480	CT image enhancement for liver cancer diagnosis
Fan et al. (2022)	Non-public (100 pairs)	Lesion image	400 × 600 × 3	Conditional entropy GAN	Adv, coe	No	SSIM PSNR	0.6182 16.7340	Cervical lesion image enhancement
Cap et al. (2023)	Non-public (2000 subjects)	Throat image	480 × 272	LaME-GAN (GAN, LaSSIM)	Adv, strp, cyc, per	Yes	LaSSIM MDOS-O*	0.936±0.038 4.05±0.12	Medical image enhancement for throat image

ing semantically meaningful features from pixel-loss images. They have already been used in image segmentation (Sect. 4.2) and classification (Sect. 4.3).

5.2 Challenges

At present, the shortcomings of current forms of GAN have been identified through limited research papers, which seriously affects their application in the medical field. Although GAN models can achieve good accuracy in medical image classification and segmentation tasks, there are still interpretability issues with the models. The models cannot accurately explain which attributes in the image region determine the classification or segmentation results, such as shape, appearance, size, or some other attribute that determines the current operation of the model. Due to the complex and variable structural characteristics of medical images, current models find it difficult to isolate the effects of individual attributes. Additionally, GAN models are typically challenging and unstable to train. The generator and discriminator are difficult to converge, and the training process is prone to problems such as gradient vanishing and mode collapse. Although most researchers proposed modified GAN architectures and customized loss functions to address these issues in the reviewed research papers, there is a lack of appropriate medical benchmarks and evaluation metrics as comparable scale standards to measure the performance of the proposed methods. For example, the best way to evaluate reconstruction results is still unclear. Finding more medically meaningful image quality evaluation metrics and more effective ways to evaluate the quality of reconstructed, synthesized, or super-resolved medical images is one of the important research directions for the future.

Most mature GAN architectures are currently 2D, and as scanning device technology updates, many medical images, such as CT and MRI, are 3D or 4D volumetric images. Therefore, it is urgent to find effective ways to proficiently apply GAN architectures in three-dimensional or even four-dimensional medical images.

To apply GANs to practical clinical medical image synthesis, analysis, and auxiliary diagnosis, some of the challenges discussed need to be solved. Therefore, we need to abandon the idea of regarding GAN as a future independent technology and instead combine it with other network models to form new hybrid network frameworks. For example, interpretable networks can be used as auxiliary modules to help improve the interpretability of GAN frameworks in medical image classification and segmentation tasks. Furthermore, we need to analyze rigorously through experiments how the GAN model framework reaches Nash equilibrium and understand its convergence in medical imaging environments, thereby solving the instability of model training. With the gradual development and satisfactory achievements in the computer vision field, this provides a good foundation for the future development of GANs. Finding image quality evaluation metrics that are more medically meaningful and universal, and more effectively evaluating the quality of synthesized or super-resolved medical images will also be one of the important research directions in the future.

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